Modeling Motorcycle Injury Severity with Southern California Crash Data

A Thesis

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Prepared By:

Amber Garcia

Ed Baek

Jeremiah Snipes

Ryan Neighbor

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Project Approval Page

We certify that we have read the project of Jeremiah Snipes, Amber Garcia, Ed Baek, Ryan Neighbor entitled MODELING MOTORCYCLE INJURY SEVERITY WITH SOUTHERN CALIFORNIA CRASH DATA and that, in our opinion, it is satisfactory in scope and quality as the thesis for the degree of Master of Science in Data Science at National University.

Approved:

_____________________________ Date
Dr. Mario Missakian
Capstone Professor, School of Technology & Engineering
National University

_____________________________ Date
Dr. Wen Cheng
Professor, Civil Engineering Department
Cal Poly Pomona

_____________________________ Date
Mr. Brian Huynh, Chief Traffic Records Officer
California Office of Traffic Safety

Dedication Page

This thesis is dedicated to the riders, to the loved ones who wait and pray for their safe return, and to those who never made it back home. Behind every crash, injury, and statistic is a life, a family, and a story that continues far beyond the roadway. May this work honor that reality in some small way by contributing to a better understanding of motorcycle injury severity and the conditions that place riders at risk. It is also dedicated to my classmates, whose shared effort and commitment helped bring this work to completion.

Acknowledgments

Use discretion in making acknowledgments. It is customary to acknowledge special assistance from extramural agencies. There is no obligation that assistance received from the faculty advisor be acknowledged. Acknowledgments should be couched in terms consistent with the scholarly nature of the work. They must be on a separate page, cannot exceed one page, and should not exceed one paragraph in length, and should adopt a restrained tone. Your name and date should not appear on this page

Abstract

The absence of structural protection places motorcyclists in direct contact with crash forces that enclosed-vehicle occupants are largely shielded from. Although motorcycles represent roughly 3% of registered vehicles in California, they account for a disproportionate share of severe and fatal traffic injuries statewide. This study examines motorcycle crash injury severity in Southern California using crash data from the California Transportation Injury Mapping System (TIMS) and registration data from the California Department of Motor Vehicles. Injury severity is modeled as a function of observable crash conditions across behavioral, environmental, roadway, spatial, and temporal domains. Logistic regression, Random Forest, and XGBoost models are applied to identify factors associated with severe and fatal outcomes. Broadside collision geometry, nighttime exposure, and alcohol involvement emerged as the strongest predictors across all models, providing an evidence base for data-driven safety interventions targeting the conditions most strongly associated with severe motorcycle injuries in the region.

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Chapter 1 Introduction

Motorcycle injury severity remains an important transportation safety and public health concern, particularly in regions where rider exposure is high and crash conditions are complex. This chapter introduces the broader problem context, defines the study motivation, presents the research questions and hypotheses, and outlines the scope and objectives guiding the analysis. Together, these elements establish the foundation for examining the factors associated with motorcycle injury severity in Southern California.

1.1 Background and Public Safety Context

Motorcycle crashes represent a persistent and disproportionate public health and transportation safety challenge in the United States. Although motorcycles comprise a relatively small share of registered vehicles and vehicle miles traveled, riders consistently account for a much larger proportion of traffic fatalities and serious injuries. (National Highway Traffic Safety Administration [NHTSA], 2024, 2025) This disparity is largely attributable to the inherent vulnerability of motorcyclists, who lack the structural protections afforded to occupants of passenger vehicles, such as seatbelts, airbags, and enclosed cabins. As a result, energy transfer during a crash is borne almost entirely by the rider's body, increasing the likelihood of severe or fatal injury. In addition to this structural vulnerability, motorcyclists do not benefit from integrated driver assistance systems commonly available in modern passenger vehicles, including lane-keeping assistance, blind-spot detection, and automated obstacle detection technologies.

Over the past two decades, these rates of injury severity in motorcyclists have remained stubbornly high despite advances in roadway engineering and vehicle safety technologies that have substantially reduced fatalities among passenger vehicle occupants. While improvements such as anti-lock braking systems (ABS) have demonstrated measurable safety benefits for motorcycles, broader safety gains have not kept pace with those observed for automobiles. (Haworth, 2012; Naweed & Blackman, 2024) This imbalance underscores the need for continued, focused investigation into the mechanisms that contribute to motorcycle crash severity.

In California, the scope of the problem is especially pronounced. The state has the largest population of registered motorcycles in the United States and consistently reports the highest number of motorcycle-related fatalities and serious injuries.(NHTSA, 2024; SafeTREC, 2025) Southern California, in particular, presents a distinctive risk environment shaped by dense urban traffic, extensive freeway networks, and a year-round riding climate that increases exposure. (SafeTREC, 2025; Mamlouk et al., 2024) Counties such as Los Angeles, San Diego, Riverside, Orange, and San Bernardino account for a substantial share of statewide motorcycle injuries and deaths, reflecting both population density and diverse roadway contexts that range from urban arterials to rural and mountainous corridors. (SafeTREC, 2025)

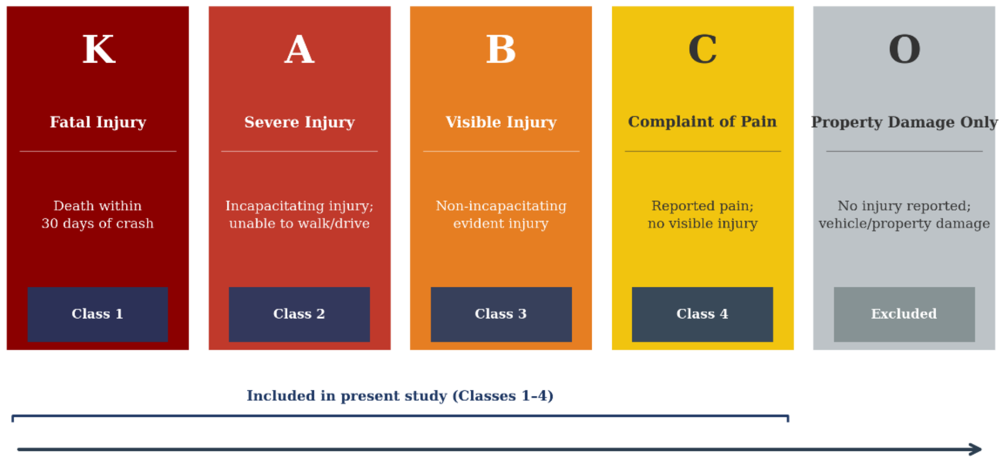
Beyond exposure, Southern California exhibits traffic characteristics that may amplify injury risk, including high-speed freeways, frequent congestion, heterogeneous vehicle mixes, and widespread use of practices such as lane-splitting. While lane-splitting is legal in California and has been studied in relation to crash frequency and injury outcomes, its interaction with speed differentials, roadway type, and rider behavior remains an area of ongoing debate.(Rice et al., 2015) These contextual factors highlight the importance of regional analyses that move beyond national averages to capture localized patterns of risk. Recent research has increasingly emphasized the multifactorial nature of motorcycle crash injury severity. Behavioral factors such as alcohol impairment, speeding, and licensing status interact with environmental conditions including lighting, weather, and roadway geometry to shape injury outcomes. (Wang et al., 2022;

Jafari et al., 2025). Moreover, temporal dimensions such as time of day and day of week have been shown to modify these relationships, with nighttime and weekend crashes often associated with higher injury severity. (Wang et al., 2022) This implies that motorcycle injury severity cannot be appropriately explained by any single factor. A better explanation comes from the interaction of behavioral, environmental, roadway, and temporal conditions.

1.2 Problem Statement and Study Motivation

Despite a breadth of existing motorcycle safety research, there are several critical gaps to be addressed. A substantial amount of the more recent prior studies focus on crash occurrence rather than injury severity. This limits insight into why some crashes result in minor injuries while others prove fatal. Among studies that do examine severity, findings are often derived from national or statewide datasets that obscure regional variation in roadway design, traffic composition, and riding culture. In California, crash injury severity is standardized using the KABCO classification system, a five-level scale ranging from fatal injury (K) through incapacitating injury (A), visible injury (B), complaint of pain (C), and property damage only (O). The present study models severity across the four injury-producing categories, excluding property-damage-only outcomes. Figure 1 illustrates the full classification structure.

Figure 1: *KABCO Classification System*



Note: KABCO classification used in California SWITRS Crash Reporting. Property-damage-only (O) excluded from modeling

Police-reported crash databases, such as those aggregated within the Transportation Injury Mapping System (TIMS), are invaluable resources for traffic safety research but present certain limitations. Key behavioral variables, including rider risk perception, aggressiveness, and situational decision-making, are rarely observed in these data. This can give rise to unobserved heterogeneity that may bias conventional modeling approaches. Additionally, inconsistencies in injury classification or under-reporting of non-fatal crashes may complicate severity analyses. In Southern California, the absence of region-specific injury severity modeling constrains the development of targeted motorcycle safety interventions. In the absence of localized evidence, policymakers and transportation agencies must often rely on generalized findings that may not fully capture the dynamics of high-density urban traffic, complex roadway configurations, or regionally prevalent riding practices. This limitation is particularly consequential given California's prominent role in motorcycle policy development, safety research, and infrastructure planning.

Subsequently, there is a need for focused investigation that identifies observable roadway, environmental, temporal, spatial, and exposure-related characteristics associated with severe and fatal motorcycle injuries in Southern California, while examining how these factors operate within complex traffic and riding contexts.

1.3 Research Questions

This study is guided by the following research questions:

1. Which observable behavioral, environmental, and roadway characteristics are associated with increased odds of severe or fatal motorcycle injury in Southern California crashes?
2. Are these injury severity associations consistent across temporal and roadway contexts, or do risk patterns differ by time of day, day of week, and roadway type?

These research questions establish the inferential focus of the study. The following hypotheses translate these questions into statistically testable statements.

1.4 Hypotheses

To formally evaluate the research questions, this study specifies hypotheses that reflect both inferential relationships and predictive performance in motorcycle injury severity modeling. Because the analysis includes multinomial logistic regression, Random Forest, and XGBoost, the hypotheses are structured not only to assess whether specific predictor domains are related to injury severity, but also whether they contribute meaningful predictive information for distinguishing among KABCO injury outcomes. For each predictor domain, the null hypothesis states that the variables do not meaningfully improve the prediction of motorcycle injury severity, whereas the alternative hypothesis states that the variables provide meaningful predictive value and help distinguish more severe outcomes from less severe outcomes.

$H0_1$ (Null Hypothesis): Roadway and traffic-environment characteristics, including intersection configuration, roadway classification, surface condition, and traffic control devices, do not provide meaningful predictive value for motorcycle injury severity outcomes.

$H1_1$ (Alternative Hypothesis): Roadway and traffic-environment characteristics provide meaningful predictive value for motorcycle injury severity outcomes and improve the model's ability to distinguish among injury severity classes.

$H0_2$ (Null Hypothesis): Environmental conditions and collision mechanisms, including lighting condition, roadway surface condition, and collision type, do not meaningfully predict motorcycle injury severity outcomes.

$H1_2$ (Alternative Hypothesis): Environmental conditions and collision mechanisms meaningfully predict motorcycle injury severity outcomes, with unlit or dark lighting conditions and broadside collision geometry contributing to increased probability of severe or fatal injury classifications.

$H0_3$ (Null Hypothesis): Temporal factors, including time of day, day of week, and weekend versus weekday periods, do not provide meaningful predictive information for motorcycle injury severity outcomes.

$H1_3$ (Alternative Hypothesis): Temporal factors provide meaningful predictive information for motorcycle injury severity outcomes and improve the model's ability to identify higher-risk injury contexts.

$H0_4$ (Null Hypothesis): Spatial context and exposure measures, including county, city, and county-level motorcycle and vehicle registration counts, do not provide meaningful predictive value for motorcycle injury severity outcomes.

$H1_4$ (Alternative Hypothesis): Spatial context and exposure measures provide meaningful predictive value for motorcycle injury severity outcomes and help explain geographic variation in severe and fatal injury classifications.

These hypotheses define the analytical structure for evaluating whether key predictor domains contribute meaningful explanatory insight and predictive usefulness in modeling motorcycle

1.5 Objectives of study

The primary objectives of this study are threefold. First, this study seeks to identify observable roadway, environmental, temporal, and exposure related characteristics that are associated with severe or fatal motorcycle injury severity outcomes in Southern California. Second, it aims to evaluate whether these associations remain stable across different temporal contexts and roadway types, or whether risk patterns will vary in meaningful ways. Third, this study intends to generate findings that can inform targeted safety interventions, policy development, and future research on high-risk conditions and populations.

1.6 Scope and Limitations

Several limitations should be acknowledged. This analysis relies on police-reported crash data, which may be subject to reporting bias, misclassification, and missing information. Certain rider-related risk factors relevant to injury severity are not directly observable within these data and must therefore be inferred through violation-based indicators and contextual modeling approaches. In addition, long-term injury outcomes are not captured in the available datasets, limiting the analysis to acute injury severity at the time of the crash. Finally, the geographic focus on Southern California may limit the generalizability of findings to regions with different traffic environments, roadway characteristics, or riding cultures.

1.7 Definition of Terms

Injury Severity: The categorical classification of crash outcomes, typically ranging from minor injury to severe injury or fatality.

Behavioral Factors: Rider-related actions or conditions observable in crash records, such as alcohol involvement or speeding.

Environmental Conditions: External factors including lighting conditions, weather, and roadway surface conditions present at the time of crash.

Temporal Context: Time-related characteristics of a crash, including time-of-day and day-of-week.

Riding Culture: The collective norms, values, and informal practices shared among motorcyclists that influence riding behavior, safety practices, and risk tolerance. Riding culture is shaped by factors such as regional norms, peer influence, riding purpose, and exposure to safety messaging, and may indirectly affect injury severity through behavioral mechanisms observable in crash data.

CRISP-DM (Cross-Industry Standard Process for Data Mining): A structured, iterative framework for organizing data science and data mining projects into six sequential phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. In this study, CRISP-DM provides the methodological scaffold guiding the progression from problem definition through model evaluation and policy interpretation.

KABCO Scale: A standardized injury severity classification system used in police-reported crash databases across the United States. The scale categorizes crash outcomes into five levels: fatal injury (K), incapacitating injury (A), non-incapacitating visible injury (B), complaint of pain (C), and property damage only (O). In this study, injury severity is modeled across the four injury-producing categories (K, A, B, C), with property-damage-only outcomes excluded from analysis.

SWITRS/TIMS (Statewide Integrated Traffic Records System / Transportation Injury Mapping System): SWITRS is California's statewide police-reported crash database, collecting standardized officer-recorded information on crash circumstances, environmental conditions, roadway characteristics, and injury outcomes for all reported traffic collisions. TIMS is the University of California Berkeley's public-facing platform that provides access to SWITRS data through a geographic interface. In this study, TIMS serves as the primary data access portal for the SWITRS crash records used in analysis.

Multinomial Classification: A supervised machine learning and statistical modeling task in which the outcome variable contains three or more unordered discrete categories. Unlike binary classification, which distinguishes between two outcomes, multinomial classification estimates the probability of membership across all outcome classes simultaneously. In this study, motorcycle injury severity is treated as a four-class multinomial outcome corresponding to the K, A, B, and C levels of the KABCO scale.

Exposure Variable: A measure that quantifies the opportunity for a crash or injury to occur, used to contextualize raw crash counts relative to the size of the at-risk population. Without exposure controls, areas with more riders will appear more dangerous simply because more crashes occur there in absolute terms. In this study, county-level motorcycle registration counts sourced from the California DMV serve as the primary exposure variable, enabling differentiation between counties with high crash frequency due to high ridership versus those exhibiting elevated crash risk relative to their registered motorcycle population.

1.8 Summary

To summarize Chapter one, motorcycle injury severity affects public health and transportation safety due to the persistent disproportionate risk faced by motorcyclists relative to other road users. The multifactorial nature of injury severity highlights how behavioral, environmental, roadway, and temporal factors interact to shape motorcycle crash outcomes. This chapter identified critical gaps in existing research, most notably in the limited availability of region-specific severity models that leverage routinely collected crash data to support targeted safety interventions. Within the context of Southern California's dense traffic environment, complex roadway infrastructure, and high motorcycle exposure, these gaps underscore the need for localized analyses that move beyond the national averages to capture distinct patterns of risk. Building on this foundation, Chapter two will situate the study within a broader body of motorcycle safety research by reviewing prior literature on injury severity, with an emphasis on behavioral, environmental, roadway, and temporal factors that motivate the analytical approach adopted in this study.

Chapter 2: Literature Review

This chapter reviews the literature relevant to motorcycle injury severity and situates the present study within the broader transportation safety research base. The review focuses on behavioral, environmental, roadway, spatial, temporal, and methodological factors that have been shown to influence injury outcomes in motorcycle crashes. Together, these studies provide the conceptual and empirical foundation for the variable selection, modeling strategy, and regional focus used in this thesis.

2.1 Global and National Vulnerability

Motorcyclists are persistently over-represented in road traffic injuries across diverse geographic and regulatory contexts. In the United States motorcyclists accounted for 15.0-15.5% of all traffic fatalities in 2022 and 2023 while only making up 3.1-3.5% of registered vehicles.(NHTSA, 2024, 2025) Internationally, where motorcycles are typically more utilitarian in nature versus the largely recreational riding culture seen in the United States, there still exists a dramatic intersection of fatality risk, injury distribution, and contributing factors. For example, in Taiwan, motorcyclists represent over 64% of all traffic fatalities and 85% of injuries

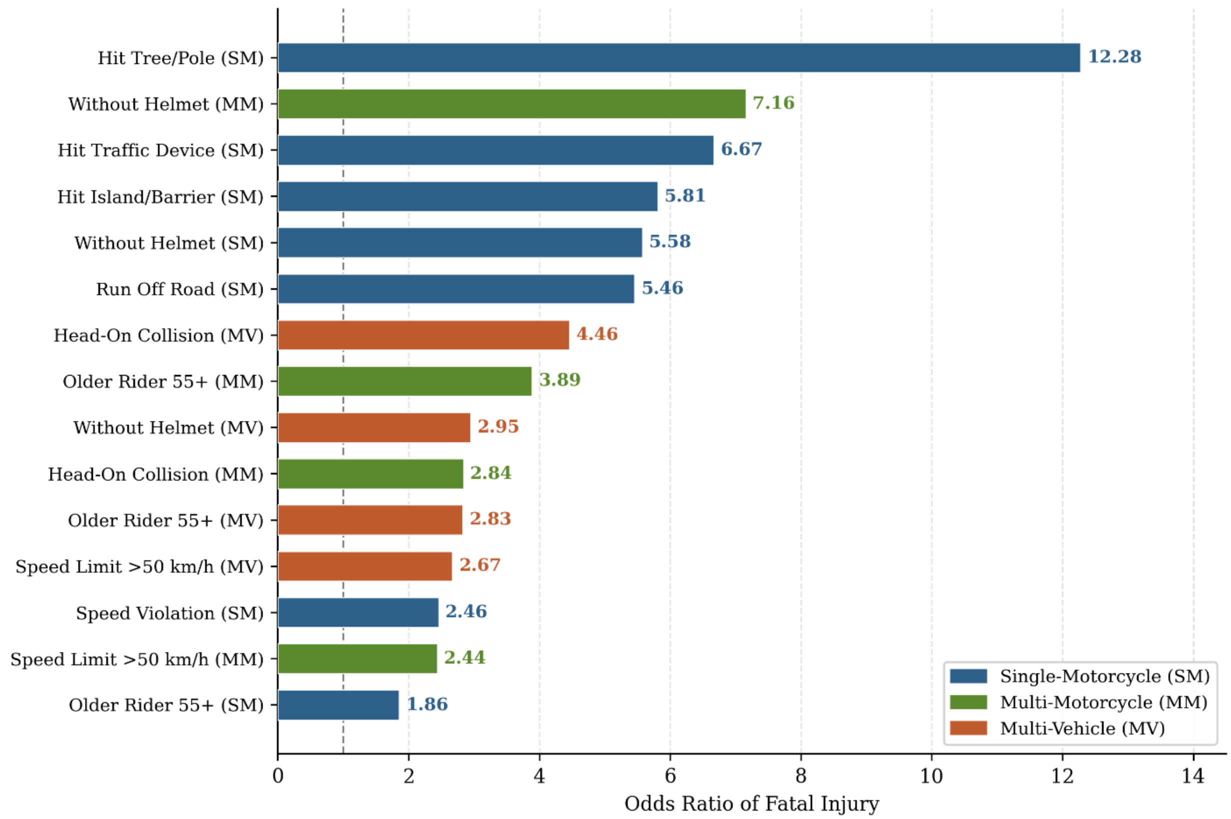
In China, 17.0-19.0% of total road collision fatalities are attributed to motorcycle crashes. Despite the wide range in diverse roadway infrastructure, enforcement practices, and riding cultures, these findings converge on the conclusion that the physical vulnerability of motorcyclists places them at elevated risk in mixed-traffic environments. Comparative analyses suggest that this vulnerability is persistent regardless of regional context, underscoring the global relevance of motorcycle injury severity as a transportation safety concern.

Recent literature also highlights inequities that are noticeably present in the distribution of safety technologies across motorcycle types, sometimes described as a “safety as luxury” phenomenon. Advanced rider assistance systems and enhanced braking or stability technologies are more commonly available on newer and higher cost motorcycles, directly limiting their availability to a subset of riders. As a result, riders operating smaller or older motorcycles often lack access to protective technologies that have become increasingly standard in passenger vehicles. This imbalance reinforces the structural disparities in crash outcomes and raises important questions about the extent to which technological advances have contributed to population level safety improvements among motorcyclists.

2.2 Crash Context and Violation-Based Risk Indicators

Prior research has consistently demonstrated that rider-related risk mechanisms contribute substantially to motorcycle injury severity, while some of these mechanisms may not be directly observable through police-reported crash data. These factors are commonly captured through violation-based indicators, such as driving under the influence, speeding, and improper turning to highlight a few. (Chang & Yeh, 2007; Wang, 2022; Naseralavi et al., 2025) These factors reflect elevated risk-taking and impaired decision-making within specific crash contexts. Figure 2 summarizes the odds ratios associated with the most influential behavioral, infrastructure, and rider characteristic predictors identified by Wang et al. (2022), illustrating the relative magnitude of each risk factor across crash configurations.

Figure 2: *Significant Risk Factors for Motorcyclist Fatality by Crash Configuration*



Note: Odds Ratios from Mixed Logit Models. BAC-unknown categories excluded due to informative missingness. Reference group= Baseline category within each crash configuration. Source: Wang et al. (2022)

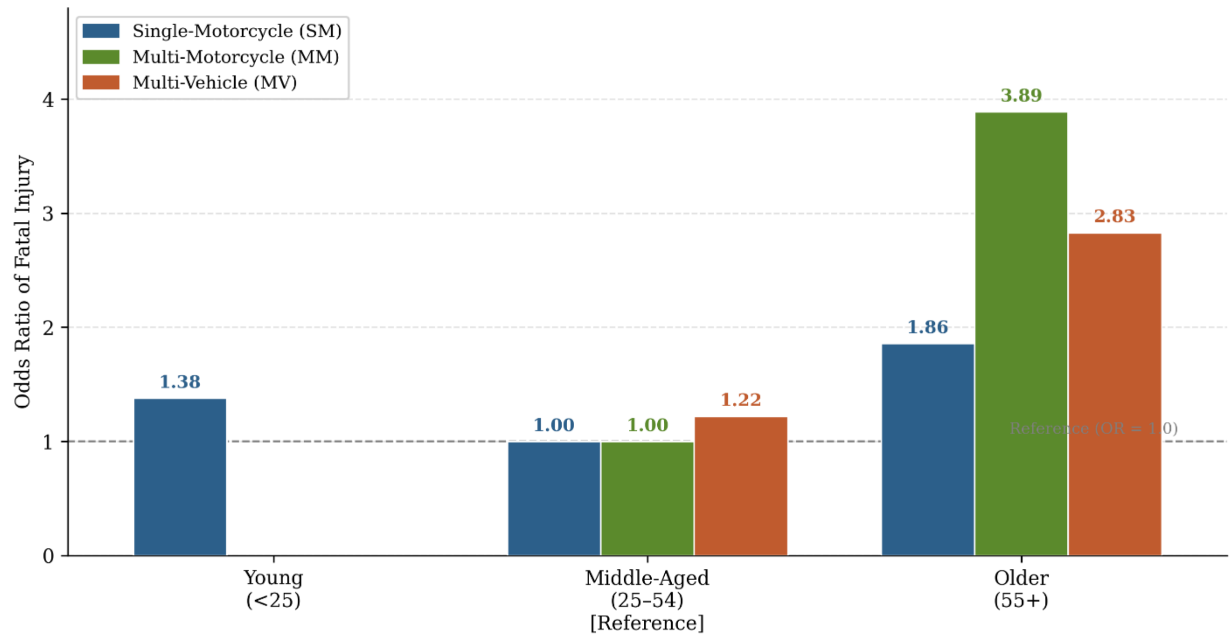
2.2.1 Alcohol Involvement in Prior Studies

Alcohol involvement remains one of the most consistent predictors of severe and fatal outcomes across literature. Elevated blood alcohol concentration levels are strongly associated with increased fatality risk across rural and urban settings (Zlatkovic & Zlatkovic, 2022; Wang, 2022). Speed-related violations represent another dominant mechanism. Excessive speed reduces available reaction time and increases kinetic energy transfer during impact, disproportionately resulting in incapacitating or fatal injuries, particularly on high-speed facilities and during low visibility conditions (Champahom et al., 2023; Farid & Ksaibati, 2021).

2.2.2 Age-related Patterns

Age-related patterns in motorcycle crashes present a more complex picture. Younger riders, especially those between 15 and 24 years-old, tend to experience higher crash involvement rates, often attributed to inexperience or thrill-seeking behavior. (Chang & Yeh, 2007; Islam & Li, 2026) However, some prior studies suggest that younger riders may be less likely to sustain fatal injuries in comparison to older riders due to greater physical resilience and faster reaction times. Older riders, typically aged 55 and above, often exhibit lower crash frequency but face significantly higher odds of severe or fatal injuries when said crashes occur. (Wang, 2022) In Taiwan, for example, riders aged 55 and above were 1.86 times more likely to be killed in single-motorcycle crashes and 3.89 times more likely in crashes that involved multiple motorcycles (Wang et al., 2022). Figure 3 illustrates how fatality odds ratios rise sharply with rider age, with older riders (55+) facing odds 1.86 to 3.89 times greater than middle-aged riders depending on crash configuration (Wang et al., 2022)

Figure 3: *Fatality Odds Ratios by Rider Age Group and Crash Configuration*



Note: Odds Ratios from Mixed Logit Models. Middle-Aged Riders (25-34) serve as the reference group. Young Rider OR reported for single-motorcycle crashes only. Source: Wang et al. (2022)

Research consistently identifies an intermix of factors that contribute to these higher fatality risks. While older riders potentially exhibit greater maturity and experience, these common traits are often over-shadowed by the natural decline in physical resilience and reaction time. A dominant theme in literature revolves around this physical decline and lessened ability to absorb the same level of kinetic energy as younger riders without sustaining life-threatening injuries. Researchers have attributed the likelihood of younger riders sustaining fatal outcomes due to sharper reflexes and faster response times. There is also the matter of unobserved heterogeneity in older riders, highlighting factors like pre-existing conditions and overall physical health status having a significant effect on their survival odds post-crash.

2.2.3 Rider Behavioral Profiles and Data Limitations

Emerging literature has additionally explored rider personality traits, distinguishing between aggressive, risk-tolerant profiles and more cautious riding styles (Haque et al., 2008).

However, these characteristics are rarely captured directly in police-reported crash databases, reinforcing the limitations of conventional datasets in fully modeling behavioral risk.

Together, these behavioral findings suggest that injury severity is not solely a function of crash occurrence, but of risk amplification mechanisms embedded within rider decision-making and demographic vulnerability.

2.3 Roadway Geometry and Infrastructure

Beyond rider behavior, research has consistently highlighted roadway and traffic environment characteristics as major determinants of motorcycle crash severity. Key predictors include high-speed road classifications, complex roadway geometry, geographic setting, intersection density, and surface conditions.

Crashes on high-speed facilities, such as interstates and principal arterials, consistently result in the highest injury severity. Elevated travel speeds reduce reaction time and substantially increase crash impact energy, a pattern supported by both national and state-level analyses (Adeyemi et al., 2023; Farid & Ksaibati, 2021).

Roadway geometry further amplifies risk. Approximately one-third of motorcycle fatalities occur on curved segments. Small curve radii and steep grades significantly increase the likelihood of loss-of-control crashes and fatal outcomes, particularly when sharp curves are combined with slopes exceeding 4% (Jafari et al., 2025; Wei et al., 2025). These geometric features reduce stability and limit corrective maneuvering time.

Geographic context introduces additional variation. Although urban areas experience higher crash frequencies due to traffic density, rural roads demonstrate greater fatality risk. This disparity is largely attributed to higher operating speeds, increased fixed-object collisions, and longer emergency response times in rural environments (Champahom et al., 2023; Li et al., 2021; Wang, 2022).

Infrastructure density also plays a meaningful role. Greater intersection and access point density per kilometer is associated with higher crash frequency and increased severity, particularly in multi-vehicle collisions (Sharafeldin et al., 2022). The presence of roadside fixed objects and guardrails further elevates fatal injury risk due to high energy transfer during impact (Ospina-Mateus et al., 2021; Santos et al., 2024).

Surface conditions significantly influence crash outcomes as well. Snowy, icy, and wet pavements compromise vehicle stability and increase the likelihood of severe injury. Pavement friction, although often under-measured in crash datasets, has been shown to reduce serious and fatal injuries at intersections and high-demand locations (Mamlouk et al., 2024; NHTSA, 2024). Collectively, these findings underscore that roadway design and infrastructure characteristics are not merely contextual factors but active determinants of injury severity.

2.4 Spatial, Exposure, and Environmental Dynamics

Spatial context and exposure intensity further shape motorcycle crash severity patterns. Roadway hierarchy remains one of the strongest predictors, with interstate and highway crashes demonstrating significantly elevated fatal injury risk across time periods. Fatal crash injuries on interstates show incident rate ratios ranging from 1.82 to 2.27 across time-of-day comparisons (Adeyemi et al., 2023). Rural roadway environments similarly demonstrate greater severity relative to urban settings (Champahom et al., 2023; Farid & Ksaibati, 2021).

2.4.1 Exposure Measures and Traffic Volume

Exposure-related measures provide additional explanatory power. Traffic volume indicators, including Annual Average Daily Traffic (AADT) and motorcycle-specific AADT, consistently predict crash frequency and severity. In machine learning applications, AADT and truck-specific AADTT frequently rank among the most influential predictors (Ospina-Mateus et al., 2021; Rezapour et al., 2020; Zlatkovic & Zlatkovic, 2022). Annual riding mileage also serves as a significant exposure metric. Studies reveal that total kilometers traveled per year strongly influence injury severity, with some research identifying a "low-mileage bias," wherein riders logging fewer than 10,000 miles annually exhibit higher crash rates (Islam & Li, 2026; Wei et al., 2024).

2.4.2 Lighting and Environmental Conditions

Environmental and lighting conditions further interact with spatial exposure. Nighttime crashes, particularly on unlit roadways, significantly increase the probability of fatal or serious injuries due to reduced visibility and delayed hazard detection (Adeyemi et al., 2023; Naseralavi et al., 2025). Crashes occurring between midnight and early morning hours (00:00-06:00) are consistently associated with the highest fatality risks. For example, research in Taiwan identified midnight travel as a critical predictor across multiple crash configurations, while a study in Mauritius reported an odds ratio of 4.787 for severe injuries during early morning hours compared to midday off-peak periods (Beeharry et al., 2021).

2.4.3 Temporal Patterns and Seasonal Trends

Temporal exposure patterns reinforce these findings. Weekend riding, particularly from Friday through Sunday, more than doubles motorcycle-related trauma admissions relative to weekdays, reflecting increased recreational riding and higher operating speeds (Smith et al., 2018; Mamlouk et al., 2024). Seasonal trends show injury peaks during summer months, especially between May and August, with some regions reporting crash frequencies exceeding annual averages by more than 60% (Naseralavi et al., 2025). Although winter months generally involve fewer crashes due to reduced exposure, adverse pavement conditions may increase minor injury likelihood among older riders (Wang, 2022).

2.4.4 Temporal Instability in Key Predictors

These temporal effects also change over time, rather than remaining consistent across years or conditions.. Recent research demonstrates considerable temporal instability in key predictors, including nighttime travel, helmet effectiveness, and speeding behaviors. In Pennsylvania, nighttime crashes were significantly associated with severe injuries until 2011, after which the relationship weakened or stabilized (Li et al., 2021). Similarly, helmet effectiveness has been shown to fluctuate across time-of-day and seasonal contexts, reinforcing the need for spatiotemporal modeling techniques capable of accounting for non-stationary relationships (Ijaz et al., 2022).

In combination, these findings illustrate that spatial context, exposure intensity, environmental conditions, and temporal instability collectively shape motorcycle crash severity. Modeling approaches must therefore account for both geographic variation and dynamic temporal effects to accurately estimate risk patterns.

2.5 Methodology and Advanced Modeling

The methodological evolution of motorcycle injury severity research reflects researchers growing recognition that crash outcomes are shaped by complex, non-linear, and context dependent interactions. Traditional discrete outcome models, including binary, ordered and multinomial logit frameworks, provided early foundations for severity analysis (Lin et al., 2003; Chang & Yeh, 2007; Hsu & Wen, 2022). However, these models assume fixed parameter effects and homogeneous relationships across observations.

Binary logit and probit models were commonly applied when outcomes were dichotomized into fatal versus non-fatal categories (Pervez et al., 2022; Wang, 2022). Although computationally efficient, these approaches reduce injury granularity and may obscure variation across intermediate severity levels. To preserve ordinal structure, ordered logit and ordered probit models were widely utilized for injury scales such as the KABCO classification system (Agyemang et al., 2021; Champahom et al., 2023). These models assume proportional odds and fixed parameter effects across observations. While statistically rigorous, they rely on the assumption that predictor effects are homogeneous across riders and crash contexts. Multinomial logit (MNL) models were introduced to accommodate multi-category injury outcomes without imposing ordinality constraints (Hsu & Wen, 2022; Zlatkovic & Zlatkovic, 2022).

However, the standard MNL framework is constrained by the Independence of Irrelevant Alternatives assumption and does not adequately account for unobserved heterogeneity. Nested logit models partially relax this assumption by grouping related severity outcomes into hierarchical structures, though they continue to rely on fixed coefficients within nests.

2.5.1 Traditional Discrete Outcome Models

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2.5.2 Heterogeneity-Sensitive and Bayesian Approaches

Recognition of unobserved heterogeneity marked a major methodological advancement. Mixed logit or Random Parameters Logit (RPL) models allow coefficients to vary across observations, reducing omitted variable bias and improving statistical fit (Farid & Ksaibati, 2021; Ijaz et al., 2022). More advanced extensions incorporate heterogeneity in means and variances, enabling covariates to explain both the central tendency and dispersion of random effects (Ijaz et al., 2022; Tian & Zhang, 2024). Latent class models provide an alternative by probabilistically segmenting crash observations into distinct behavioral profiles (Islam & Li, 2026). Bayesian multivariate frameworks further enhance estimation by incorporating hierarchical structure and prior distributions to address temporal and multilevel correlations
(Cheng et al., 2017).

2.5.3 Spatiotemporal Modeling

Recent studies have also emphasized spatial and temporal non-stationarity. Spatiotemporal approaches such as geographically weighted and temporally dynamic models allow coefficients to vary across location and time, improving localized inference and predictive accuracy (Li et al., 2021; Adeyemi et al., 2023; Jafari et al., 2025). These models outperform global fixed-parameter approaches by identifying regional clusters and evolving risk patterns.

2.5.4 Machine Learning and Explainable AI

The most recent phase of methodological development incorporates machine learning and deep learning techniques aimed at enhancing predictive performance and modeling nonlinear interactions. Decision trees and ensemble approaches such as Random Forest have been widely applied to rank variable importance and reduce overfitting (Rezapour et al., 2021; Santos et al., 2021). Support Vector Machines and Multivariate Adaptive Regression Splines have also demonstrated competitive classification performance relative to traditional regression models (Wahab & Jiang, 2019). Deep learning architectures, including artificial neural networks and related frameworks, have further improved prediction accuracy in injury severity modeling contexts (Rezapour et al., 2020). Since machine learning models are frequently criticized for limited interpretability, recent studies have incorporated explainable artificial intelligence techniques to enhance transparency. Interpretable machine learning frameworks, including SHAP-based analyses, allow researchers to quantify the marginal contribution of individual predictors to injury severity outcomes (Wei et al., 2024). Across modeling approaches, performance is commonly evaluated using log-likelihood statistics, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and pseudo-R^2 metrics to assess model fit and transferability (Farid & Ksaibati, 2021; Li et al., 2021). Over time, motorcycle injury severity research has shifted from fixed parametric models toward more flexible approaches that account for heterogeneity and nonlinear relationships.

2.6 Summary and Identified Gaps

The literature reviewed in this chapter demonstrates that motorcycle injury severity is shaped by a complex interplay of rider characteristics, behavioral risk indicators, roadway design, environmental conditions, spatial exposure, and temporal dynamics. Across diverse geographic contexts, motorcyclists remain disproportionately represented in severe and fatal traffic injuries, underscoring the structural vulnerability inherent in two-wheeled transportation (NHTSA, 2025).

Behavioral indicators such as alcohol impairment, speeding, and risky maneuvering consistently emerge as dominant predictors of severe outcomes (Chang & Yeh, 2007; Wang, 2022; Naseralavi et al., 2025; Zlatkovic & Zlatkovic, 2022). Age-related patterns reveal differential vulnerability across the life course, with younger riders exhibiting higher crash involvement and older riders experiencing greater fatality risk once a crash occurs (Islam & Li, 2026; Wang, 2022). These findings reinforce the importance of accounting for both exposure intensity and physiological resilience when modeling severity (Islam & Li, 2026; Wang, 2022).

Roadway geometry and infrastructure further amplify injury risk. High-speed facilities, sharp curves, steep grades, intersection density, and fixed-object collisions consistently increase severity probabilities (Adeyemi et al., 2023; Farid & Ksaibati, 2021; Jafari et al., 2025; Wei et al., 2025; Sharafeldin et al., 2022). Environmental and temporal factors, including nighttime travel, seasonal riding patterns, and weekend recreational exposure, interact with roadway context to intensify injury outcomes (Naseralavi et al., 2025; Smith et al., 2018; Li et al., 2021). Importantly, multiple studies demonstrate spatial and temporal instability in predictor effects, indicating that relationships between risk factors and injury severity are not constant across time or location (Li et al., 2021; Ijaz et al., 2022; Jafari et al., 2025). A comprehensive summary of statistically significant predictors, effect sizes, and odds ratios identified across the studies reviewed is provided in Appendix A, offering a cross-study reference for the variable selection decisions discussed in Chapter 3.

Methodologically, the field has evolved from traditional discrete outcome models toward heterogeneity-sensitive and spatiotemporal frameworks (Lin et al., 2003; Chang & Yeh, 2007; Farid & Ksaibati, 2021). Random parameter models, latent class approaches, Bayesian methods, and geographically weighted techniques have been developed to address unobserved heterogeneity and non-stationarity (Cheng et al., 2017; Ijaz et al., 2022; Tian & Zhang, 2024; Li et al., 2021). More recently, machine learning and interpretable artificial intelligence techniques have expanded predictive capacity while attempting to preserve explanatory transparency (Rezapour et al., 2020; Santos et al., 2021; Wei et al., 2024).

Despite these advances, persistent gaps remain. Many commonly used datasets lack detailed measures of rider behavior, exposure intensity, roadway micro-engineering characteristics, and protective equipment quality (Islam & Li, 2026; Sharafeldin et al., 2022; Burford et al., 2024). Spatial instability is frequently under-modeled, and exposure measures such as annual riding mileage are inconsistently incorporated (Islam & Li, 2026; Li et al., 2021). These limitations constrain the precision of severity estimation and may obscure context-specific risk mechanisms.

Across prior literature, motorcycle injury severity is consistently shown to be dynamic, multifactorial, and context dependent. Accordingly, the present study adopts a region-specific modeling strategy that integrates logistic regression and machine learning approaches to evaluate injury severity within Southern California's distinct spatial and exposure context. Chapter 3 outlines the methodological framework, including data sources, variable operationalization, model specification, and validation procedures designed to address the limitations identified in prior research.

Chapter 3: Methodology

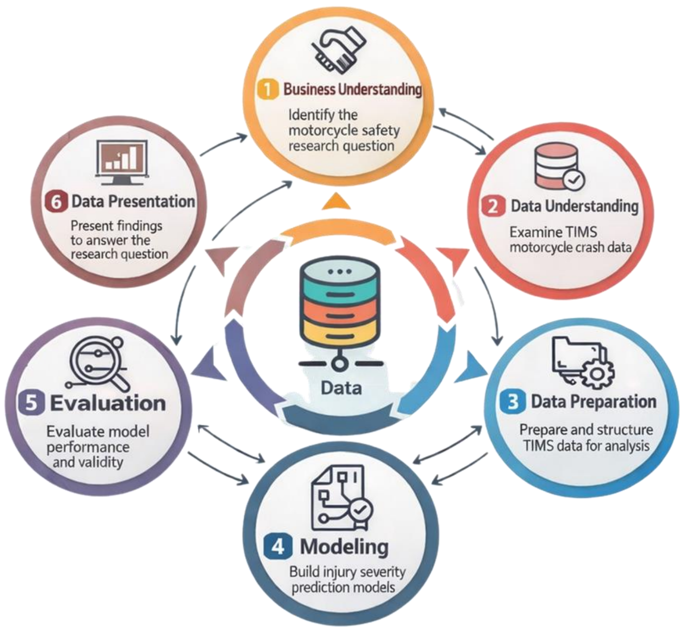
This chapter describes the methodological framework used to examine motorcycle injury severity in Southern California. It outlines the study design, data sources, variable definitions, preparation procedures, modeling strategy, and evaluation methods used to support both explanatory analysis and predictive comparison. Together, these components establish the analytical structure used to identify the factors associated with injury severity outcomes.

3.1 Study Design and Analytical Framework

This study employs a retrospective observational research design using secondary police-reported crash data to examine motorcycle injury severity outcomes in Southern California. The unit of analysis is the individual motorcycle-involved crash event. Because the dataset consists of historical administrative records collected through standardized reporting systems, no experimental manipulation or intervention is present. All inferences are derived from observable roadway, environmental, temporal, behavioral, and exposure-related characteristics measured at the time of the crash. The study is both explanatory and predictive in nature: it evaluates statistically significant associations between crash conditions and injury severity while also assessing the classification performance of competing modeling frameworks.

The analytical workflow is guided by the Cross-Industry Standard Process for Data Mining (CRISP-DM). This framework structures the study into sequential and iterative phases that align with the organization of the thesis. Chapter 1 establishes the business understanding phase by defining the public safety problem, research purpose, and policy significance. Chapter 2 supports data understanding conceptually by synthesizing prior research on behavioral, environmental, roadway, and temporal determinants of injury severity. Chapter 3 operationalizes the data understanding and preparation phases by detailing data sources, variable definitions, harmonization procedures, encoding strategies, and structured integration of crash and exposure datasets.

Figure 4: *CRISP-DM Framework Guiding the Analytical Workflow*



Note. The CRISP-DM framework structures the study from business understanding through evaluation and policy interpretation.

The modeling phase is presented in Sections 3.6 and 3.7, where multinomial logistic regression, Random Forest, and Extreme Gradient Boosting are implemented and evaluated. Chapter 4 corresponds to the evaluation phase, presenting descriptive statistics, model performance metrics, comparative interpretation, and robustness checks. Finally, Chapter 5 reflects the interpretation and application phase, translating empirical findings into public safety implications, policy considerations, and recommendations for future research.

Within this framework, motorcycle injury severity modeling is treated as a supervised multiclass classification problem. Although the KABCO injury scale is ordinal, severity is modeled as a multinomial categorical outcome to avoid imposing proportional odds assumptions and to allow category-specific effects to vary independently across fatal, severe, visible injury, and complaint-of-pain outcomes. The modeling strategy integrates parametric statistical inference and ensemble machine learning to balance interpretability and predictive discrimination. Together, this structured design ensures methodological transparency, reproducibility, and alignment between research objectives, empirical analysis, and applied safety outcomes.

3.2 Data Sources

This study utilizes multiple data sources to analyze motorcycle crash severity in California comprehensively. The primary dataset, the Statewide Integrated Traffic Records System (SWITRS), provides detailed officer-reported crash records essential for modeling severity. This dataset encompasses environmental conditions, roadway characteristics, behavioral factors, and injury outcomes, thus offering a full spectrum of crash circumstances.

To contextualize these crash frequencies, the research includes annual motorcycle registration data from the California Department of Motor Vehicles (DMV). These registration counts serve as indicators of population-level exposure, assisting in differentiating high-crash counties based on ridership levels versus those exhibiting increased risk despite lower ridership.

Additional environmental data, such as weather conditions and roadway surface characteristics, enrich the analysis. By linking SWITRS data with sources that address temporal influences, like precipitation and visibility, researchers obtain a clearer understanding of variables that exacerbate crash severity. The integration of these datasets ensures a robust analytical framework spanning multiple years and all counties in California, with reliability bolstered by standardized reporting protocols.

3.3 TIMS / SWITRS Crash Data

The SWITRS dataset serves as the foundation for this study, detailing all motorcycle-related crashes in California. It includes several critical variables:

Temporal variables: COLLISION_TIME, DAY_OF_WEEK, TIME_OF_DAY

Environmental conditions: WEATHER_1, LIGHTING, ROAD_SURFACE

Roadway characteristics: CHP_ROAD_TYPE, LOCATION_TYPE, INTERSECTION

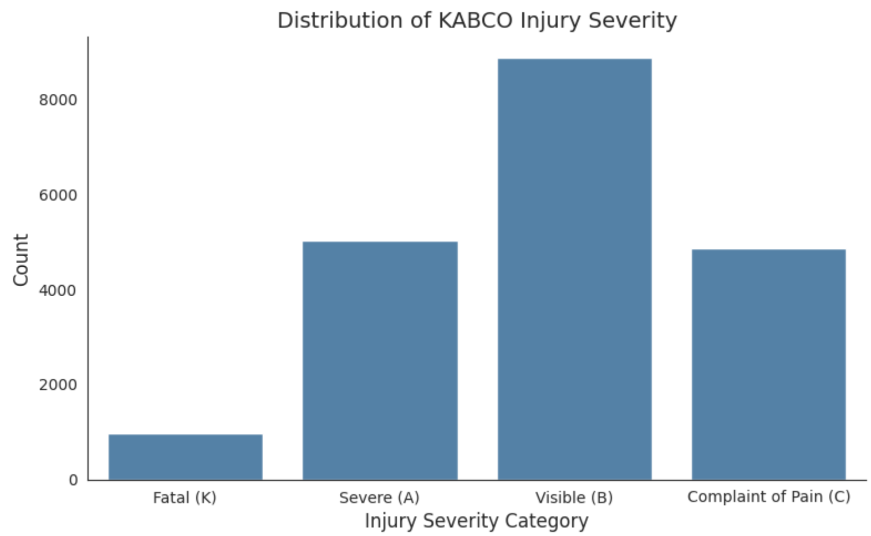
Behavioral indicators: PRIMARY_COLL_FACTOR, PCF_VIOL_CATEGORY, ALCOHOL_INVOLVED

Geospatial coordinates: LATITUDE, LONGITUDE

The key outcome variable, COLLISION_SEVERITY is categorized using the KABCO scale, which classifies crashes into fatal, severe, visible injury, and complaint of pain. Severity distributions from the dataset reveal that 4.9% were fatal, 25.5% severe injuries, 44.9% visible injuries, and 24.6% complaints of pain.

Figure 5 illustrates the distribution of injury severity categories within the analytical dataset and highlights the class imbalance present across KABCO levels.

Figure 5: *Distribution of KABCO Injury Severity*



Note. Injury severity categorized using the KABCO classification system (K, A, B, C).

Known limitations, such as missing data in certain fields and inconsistent reporting of weather conditions, were mitigated through structured missing-data strategies.

3.4 California DMV Motorcycle Registration Data

The DMV registration data provide annual counts of registered motorcycles across California counties. These counts act as exposure measures, allowing the study to assess crash frequencies relative to the number of vehicles in use. Data cleaning processes included removing unnamed columns, standardizing field names, and aligning county and year fields with SWITRS records. A left join on County x Year during merging ensured accurate exposure contexts for each crash.

The resulting exposure variables, which include Motorcycle Registration and Total Registrations help differentiate between high-crash counties due to high ridership versus those marked by heightened risk.

3.5 Supplemental Weather and Environmental Data

Environmental conditions affecting crash severity were primarily derived from SWITRS fields. Key variables include:

PRIMARY WEATHER CONDITION: WEATHER_1

LIGHTING conditions: daylight, dark with streetlights, dark without streetlights

ROAD SURFACE conditions: dry, wet, icy, or snowy

ROAD CONDITION: including debris, oil, or standing water

These variables are an important focal point, as studies have shown that factors like wet or icy pavement impair vehicle stability, while inadequate lighting heightens risks of fatal collisions. Whenever necessary, environmental variables were harmonized across years to ensure consistent data integration.

3.6 Study Population & Inclusion Criteria:

The study population comprises all recorded motorcycle crashes within SWITRS for the selected years. Inclusion criteria specify that crashes must involve at least one motorcycle, possess a valid KABCO injury severity code, occur within a California county with DMV data, and include comprehensive temporal and environmental information. Exclusion criteria eliminated property damage only crashes and any records lacking essential identifiers or suffering from data corruption. The final analytical sample includes all eligible motorcycle-related crashes with complete predictor values.

3.7 Variable Definitions

The primary outcome variable in this study is injury severity, classified using the KABCO scale. Injury severity was modeled as a four-category multinomial outcome to preserve categorical distinction across levels of injury. Although the KABCO system is ordinal in structure, severity was treated as a multinomial outcome to avoid imposing proportional odds constraints and to allow category-specific effects to vary independently across predictors.

Table 1 presents the KABCO injury severity categories used in this analysis and their corresponding modeling treatment.

Table 1: *KABCO Injury Severity Classification Used in Modeling*

Severity Level	KABCO Code	Description	Modeling Treatment
Fatal Injury	K	Death within 30 days of crash	Multinomial Class 1
Severe Injury	A	Incapacitating injury	Multinomial Class 2
Visible Injury	*B*	Non-incapacitating evident injury	Multinomial Class 3
Complaint of Pain	C	Reported pain without visible injury	Multinomial Class 4

Note. Property-damage-only (KABCO O) crashes were excluded because the study models injury severity among injury-producing motorcycle crashes.

Property-damage-only crashes (KABCO O) were excluded from the analytical sample to focus the modeling framework specifically on injury-producing crashes. This exclusion ensures alignment with the study's objective of identifying factors associated with severe and fatal injury outcomes.

Predictor variables were grouped into five conceptual domains based on theoretical grounding in prior injury severity literature and data availability within the SWITRS and DMV datasets. These domains reflect the multifactorial structure of motorcycle crash risk and include temporal, environmental, roadway, behavioral, and spatial or exposure-related factors.

Table 2 summarizes the predictor variables included within each conceptual domain and describes their operationalization for modeling purposes.

Table 2: *Predictor Variables by Conceptual Domain*

Domain	Variables Included	Operational Notes
Temporal	COLLISION_TIME; DAY_OF_WEEK; TIME_OF_DAY	Encoded into categorical bins
Environmental	WEATHER_1; LIGHTING; ROAD_SURFACE; ROAD_COND_1	Harmonized across reporting years
Roadway	CHP_ROAD_TYPE; INTERSECTION; LOCATION_TYPE	One-hot encoded
Behavioral	PRIMARY_COLL_FACTOR; PCF_VIOL_CATEGORY; ALCOHOL_INVOLVED	Violation-based indicators
Spatial and Exposure	COUNTY; LATITUDE; LONGITUDE; MOTORCYCLE_REGISTRATIONS; TOTAL_REGISTRATIONS	Merged via County x Year

Note. Predictor variables were harmonized across reporting years, encoded as appropriate, and screened to prevent outcome leakage.

Variables were harmonized across reporting years to ensure consistency. Categorical predictors were encoded using one-hot or domain-appropriate binning strategies. Variables that could introduce outcome leakage, such as direct counts of fatalities or injuries, were excluded to preserve model validity. Additionally, redundant variables and predictors exhibiting extreme sparsity were removed to improve stability and generalization performance.

3.8 Data Preparation and Integration

The data preparation phase transformed raw SWITRS crash records and county-level DMV registration data into a structured crash-level modeling dataset suitable for statistical and machine learning analysis. Preparation procedures were guided by the CRISP-DM framework, emphasizing data quality, structural consistency, and prevention of outcome leakage prior to model estimation.

3.8.1 Dataset Cleaning and Standardization

Data cleaning included standardizing column names across datasets, removing empty or redundant fields, and eliminating variables that could introduce post-outcome information (e.g., injury count fields). Sparse or proxy control variables were excluded to improve model interpretability and reduce noise.

County-level motorcycle registration data were merged with crash-level records using a left join on County and Year. This strategy preserved all motorcycle crash observations while incorporating exposure-related predictors. Following integration, the final dataset consisted of 19,721 crash-level records structured for supervised classification modeling.

3.8.2 Missing Data Treatment

Missing data in the SWITRS crash dataset does not occur randomly but instead reflects structured reporting practices and crash circumstances. For this reason, a domain-informed strategy was implemented rather than relying on listwise deletion or purely mechanical imputation procedures.

Figure 2 summarizes the extent of missingness across key predictors in the analytical dataset. The variables exhibiting the highest levels of missingness include RAMP_INTERSECTION, ALCOHOL_INVOLVED, LOCATION_TYPE, and PCF_VIOL_SUBSECTION. Importantly, the absence of values in these fields is tied to reporting mechanisms rather than data corruption. For example, ramp indicators are only recorded when a crash occurs near a ramp facility, alcohol involvement may be left unrecorded unless testing is performed, and PCF subsections are only present when a definable violation subsection applies. Consequently, missingness in these variables carries substantive meaning.

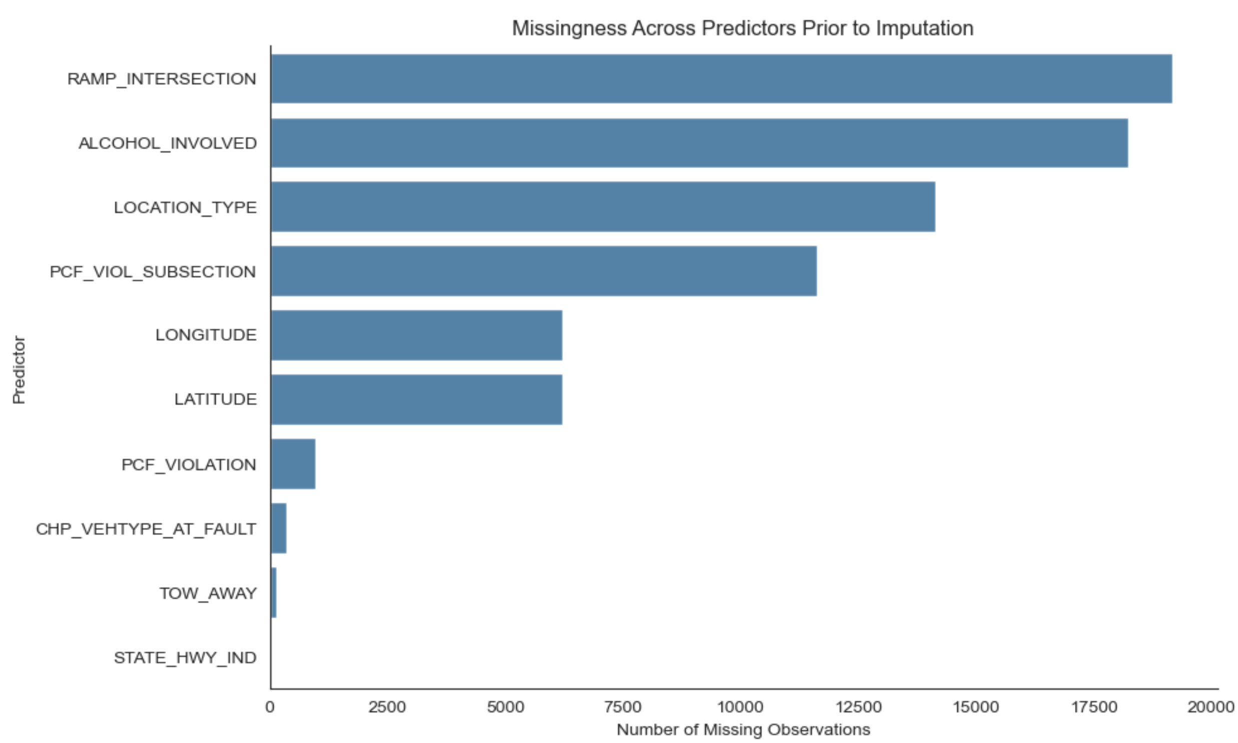
For categorical variables with significant structured missingness, such as ALCOHOL_INVOLVED and LOCATION_TYPE, an explicit "Unknown" category was created. This approach preserves the informational content of uncertainty or non-applicability while avoiding the bias that would be introduced through case deletion.

Geospatial variables (LATITUDE and LONGITUDE) exhibited moderate missingness attributable to geocoding failures rather than conceptual absence. Because these missing values do not reflect meaningful crash characteristics, median imputation was applied. This strategy preserves spatial structure without disproportionately influencing central tendency.

Categorical predictors with low levels of missingness were treated using mode imputation to maintain distributional stability without materially altering variable structure.

Collectively, this structured missing-data approach preserves interpretability, minimizes bias, and ensures analytical robustness while retaining the full crash-level sample for modeling.

Figure 6: *Missingness Across Predictors Prior to Imputation*



Note. Bars represent the number of missing observations before structured categorical encoding and median imputation were applied.

3.8.3 Final Analytical Dataset

The final analytical dataset consisted of 19,721 crash-level motorcycle observations across five Southern California counties. All crash records were preserved during data preparation and integration, as a left join strategy ensured that each motorcycle crash remained the unit of analysis.

Following removal of leakage-prone variables, sparse fields, and proxy controls, as well as low correlation variables, 2 datasets were formed. One included 16 predictor variables and a Pearson's R cutoff greater than 0.075, while the other had 8 predictor variables and a Pearson's R cutoff greater than 0.1. The fully trimmed variable clash definition can be found in **Appendix B: Model Variable Dictionary.** Both datasets' models have outcome variables of (COLLISION_SEVERITY via KABCO scale) retained for modeling. The dataset is structured such that each row represents a single crash event, with all predictor variables measured prior to the injury outcome.

Table 3: *Final Analytical Dataset Structure*

Component	Description
Total crash records retained	19,721
Predictor variables retained	16; 8
Outcome variable	COLLISION_SEVERITY (KABCO)
Unit of analysis	Individual crash

Note. All crash-level observations were preserved following data cleaning and integration. No staged exclusions were applied. The dataset was structured for supervised classification modeling.

This structured preparation ensures alignment between predictor domains and the outcome definition, supports model interpretability, and maintains full reproducibility of the supervised classification framework.

Table 4: *Predictor Variable Meanings and Classes*

Variable	Meaning	Parent Category	Full Classes
TYPE_OF_COLLISION_B	Broadside	Collision Type	A–J
TYPE_OF_COLLISION_D	Sideswipe same direction	Collision Type	A–J
TYPE_OF_COLLISION_E	Sideswipe opposite direction	Collision Type	A–J
PCF_VIOL_CATEGORY_01	DUI	PCF Category	01–20
PCF_VIOL_CATEGORY_07	Unsafe speed	PCF Category	01–20
PCF_VIOLATION_21658.0	Unsafe lane change	CVC Code	Hundreds of codes
PCF_VIOLATION_23152.0	DUI alcohol	CVC Code	Hundreds of codes
TIME_OF_DAY_Night	Day/Night	Time of Day	Morning/Afternoon/Evening/Night
ALCOHOL_INVOLVED_Y	Alcohol involved	Alcohol Indicator	Y/blank
TOW_AWAY_Y	Vehicle towed	Tow Indicator	Y/blank
LIGHTING_A	Daylight	Lighting	A–G
LIGHTING_D	Dark, streetlights on	Lighting	A–G
HIT_AND_RUN_M	Misdemeanor hit & run	Hit & Run	F/M/blank
CITY_UNINCORPORATED	Unincorporated area	City	City names + Unincorporated
STWD_VEHTYPE_AT_FAULT_C	Motorcycle at fault	Standardized Vehicle Type	A–J
CHP_VEHTYPE_AT_FAULT_02	Passenger car at fault	CHP Vehicle Type	01–99

3.9 Model Approach

Injury severity was modeled as a four-class multinomial outcome under the KABCO classification system: fatal injury (K), severe injury (A), visible injury (B), and complaint of pain (C). Although the KABCO scale is ordinal in structure, this study treats severity as a multinomial categorical outcome to avoid imposing proportional odds assumptions and to allow each severity level to vary independently across predictors. Prior injury severity research has shown that proportionality constraints may mask heterogeneous effects across severity categories.

To balance interpretability and predictive performance, three complementary modeling frameworks were implemented: multinomial logistic regression, Random Forest, and Extreme Gradient Boosting (XGBoost). These frameworks represent a methodological progression from parametric statistical inference to flexible nonlinear ensemble learning. Table 5 summarizes the structural characteristics, strengths, limitations, and analytical role of each model within the study.

Table 5: *Modeling Framework Comparison*

Model	Model Type	Key Strengths	Primary Limitations	Role In Study
Multinomial Logistic Regression	Parametric statistical model	Interpretable coefficients; statistical inference; odds ratios	Assumes linearity in log-odds; limited nonlinear interaction capture	Baseline interpretable model
Random Forest	Ensemble tree-based model	Captures nonlinear relationships; robust to overfitting; variable importance ranking	Reduced interpretability compared to regression	Nonlinear structure detection
Extreme Gradient Boosting (XGBoost)	Gradient-boosted ensemble	High predictive accuracy; handles complex interactions; regularization controls	Computationally intensive; lower transparency	Predictive performance optimization

Note. All models were trained using identical predictor variables and evaluated using macro-averaged F1, weighted F1, and ROC-AUC metrics.

The inclusion of multiple modeling frameworks enables both explanatory analysis and predictive comparison. Logistic regression provides interpretable coefficient estimates and statistical inference, while tree-based ensemble methods capture nonlinear interactions and complex feature relationships that may not be detected under parametric assumptions. Together, these models support robust evaluation of injury severity patterns across behavioral, environmental, roadway, and exposure domains.

3.9.1 Multinomial Logistic Regression

Multinomial logistic regression serves as the parametric baseline model. This framework estimates class-specific log-odds of injury severity relative to a reference category, allowing direct interpretation of predictor effects. Coefficients represent the change in log-odds associated with a one-unit increase in a predictor while holding other variables constant.

The primary strength of multinomial logistic regression lies in interpretability. Because the research objective includes identifying observable crash conditions associated with severe and fatal outcomes, a transparent statistical framework is necessary to quantify directional effects and assess statistical significance.

Given the high-dimensional feature space created through one-hot encoding of categorical variables, regularization was applied to stabilize coefficient estimation and reduce sensitivity to multicollinearity. This approach preserves interpretability while improving generalization performance.

3.9.2 Random Forest

Random Forest was selected to model nonlinear relationships and high-order interactions among roadway, environmental, temporal, behavioral, and exposure-related predictors. As an ensemble of bootstrap-aggregated decision trees, Random Forest reduces variance through averaging and mitigates overfitting relative to individual tree-based models.

Unlike logistic regression, Random Forest does not impose linearity or additivity assumptions. This flexibility is particularly important in injury severity modeling, where complex interactions may exist between lighting conditions, roadway type, temporal context, and behavioral factors.

In addition to classification performance, Random Forest provides variable importance rankings, offering insight into which predictors most strongly influence severity outcomes without requiring parametric assumptions. This supports comparative interpretation across modeling frameworks.

3.9.3 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) was implemented as a gradient-boosted ensemble model designed to optimize predictive discrimination. Unlike Random Forest, which builds trees independently, XGBoost constructs trees sequentially. Each new tree is trained to minimize residual error from prior iterations using gradient descent optimization.

This sequential learning structure enables XGBoost to capture complex nonlinear patterns and interaction effects with high predictive accuracy. Regularization parameters, learning rate controls, and tree depth constraints were incorporated to limit model complexity and prevent overfitting.

XGBoost is particularly well-suited for structured tabular datasets with heterogeneous feature domains, such as crash severity data. Its ability to balance bias and variance through controlled boosting makes it an effective comparator to both logistic regression and Random Forest.

Across all three frameworks, class-weight adjustments were incorporated to address the distributional imbalance inherent in fatal and severe injury outcomes. Because false negatives for fatal injuries carry greater public safety consequences than false positives, model optimization prioritized balanced class performance rather than overall accuracy alone.

The integration of these three models allows direct comparison between interpretable statistical inference and high-performance machine learning classification, supporting both explanatory insight and predictive robustness in evaluating motorcycle injury severity. To examine predictor influence under nonlinear ensemble modeling, feature importance rankings were extracted from the best-performing model. These rankings are presented and interpreted in Chapter 4 alongside comparative model performance results.

3.10 Model Training and Validation

This section outlines the framework used to train, tune, and evaluate the three predictive models in this study. The design prioritizes generalizability over training-set performance, and each decision regarding data partitioning, overfitting controls, and metric selection was driven by the class imbalance inherent in the KABCO severity outcome and the applied public safety objectives of the research.

3.10.1 Train-Test-Validation Split Strategy

In the first stage, 15 percent of the full encoded dataset was withheld as a held-out test set. This partition was drawn prior to any model fitting or hyperparameter selection so that the test set remained completely isolated from all training decisions. The remaining 85 percent of records were divided into a 70 percent training set and a 15 percent validation set, resulting in an overall 70/15/15 split.

This three-way split strategy was selected instead of a simple 80/20 partition because the validation set allows for independent performance monitoring during model development. Modeling decisions, including class weighting strategies, regularization settings, tree depth constraints, and learning rate parameters, were informed using validation performance. The held-out test set was reserved exclusively for final model evaluation and provides an unbiased estimate of each model's generalization capacity. Table 5 summarizes the dataset partition structure.

Table 6: *Dataset Partition Structure*

Dataset Partition	Percentage Of Total Dataset	Purpose
Training Set	70%	Model fitting and parameter estimation
Validation Set	15%	Hyperparameter tuning and model selection
Test Set	15%	Final out-of-sample performance evaluation

Note. The test set was isolated prior to any model fitting to ensure unbiased generalization assessment. Class-weight adjustments were applied during training to address KABCO severity class imbalance.

The four severity classes under the KABCO classification system exhibit distributional asymmetry, particularly the underrepresentation of fatal outcomes. To address this imbalance, class-weight adjustments were incorporated during training. These adjustments preserved natural class proportions while modifying the loss function to reduce majority-class dominance and improve sensitivity to severe and fatal injury outcomes.

3.10.2 Cross-Validation Approach

Given the scale of the dataset, the three-way holdout design provides stable and precise performance estimates comparable and the validation partition functioned as an internal monitoring throughout development. For the Random Forest, a maximum tree depth of 10 and 200 estimators were selected after observing that deeper configurations widened the training-to-validation performance gap, a diagnostic signature of overfitting. For logistic regression, convergence across the high dimensional one-hot encoded feature space. XGBoost learning rate and subsampling parameters were finalized against validation loss trajectories.

3.10.3 Overfitting Prevention

Overfitting happens when a model captures noise in training and to prevent overfitting, for logistic regression, coefficient shrinkage reduces sensitivity to predictors in encoded features. For the Random Forest, a max_depth of 10 limits and bootstrap aggregation across 200 trees reduces prediction variance through ensemble averaging. XGBoost controls complexity through its sequential shallow-tree architecture, penalize leaf-node proliferation and constrain the contribution of each successive tree. Across all three frameworks, class_weight balancing ensures that fatal and severe outcomes exert proportionally stronger influence on the objective, preventing the optimizer from defaulting to majority-class predictions.

3.10.4 Performance Metrics

Precision measures the proportion of positive predictions for a given class that are correct. Recall measures the proportion of true positives that the model identifies, and carries particular importance for the fatal injury class, where false negatives carry greater public safety consequences than false positives. The F1 score is the mean of precision and recall, balancing both. Macro averaged F1 treats all four severity classes equally regardless of frequency, giving proportionally greater weight to underrepresented classes such as fatal injury, and serves as the primary comparative metric across models. Weighted average F1 scales each class score by its sample frequency, reflecting aggregate performance across the full severity distribution. ROC-

AUC is generating per-class, micro-averaged, and macro-averaged statistics as measure ranging from 0.5 to 1.0 or chance to perfect discrimination, ROC-AUC enables direct comparison of probabilistic discrimination across all three models.

3.10 Limitations in Model Design

Although the modeling framework was designed to maximize predictive robustness and methodological transparency, several limitations inherent to secondary crash data and supervised classification modeling must be acknowledged. First, the analysis relies on police-reported administrative records, which may contain reporting inconsistencies, classification variability, and measurement error in behavioral and environmental variables. Injury severity coding under the KABCO scale is subject to on-scene assessment and may not perfectly reflect clinical outcomes.

Second, the observational nature of the dataset limits causal inference. While logistic regression provides adjusted association estimates and ensemble models capture nonlinear interactions, all results should be interpreted as associative and predictive rather than causal. Unmeasured confounding factors such as rider experience, helmet quality, emergency response time, and post-crash medical care are not fully captured in the dataset and may influence injury severity outcomes.

Third, class imbalance within the KABCO outcome distribution introduces modeling complexity. Although class-weight adjustments and evaluation metrics such as macro-averaged F1 were implemented to mitigate majority-class dominance, rare fatal outcomes remain statistically challenging to model with high precision.

Fourth, one-hot encoding of categorical variables expands dimensionality and may reduce interpretability for tree-based importance rankings. While regularization and depth constraints were applied to control overfitting, high-dimensional feature space inherently increases variance sensitivity.

Finally, model performance is evaluated using structured holdout partitions within Southern California counties and may not generalize to other geographic regions without recalibration. Roadway infrastructure, traffic patterns, enforcement practices, and environmental exposure vary across jurisdictions.

These limitations do not invalidate the findings but frame their appropriate interpretation within predictive modeling rather than causal policy experimentation.

3.11 Conclusion

This methodological framework establishes a structured and reproducible approach for modeling motorcycle injury severity using integrated administrative crash and exposure datasets. Guided by the CRISP-DM process, the study systematically progresses from problem definition and data harmonization through supervised multiclass classification and comparative model evaluation.

By combining parametric statistical inference with ensemble machine learning, the design balances interpretability and predictive performance. Logistic regression provides transparent effect estimation, while Random Forest and XGBoost capture nonlinear relationships and interaction effects across behavioral, environmental, roadway, and exposure domains.

The implementation of a three-way train–validation–test split, class-weight adjustments, and performance metrics aligned with public safety priorities ensures that model evaluation emphasizes generalizability and balanced class performance rather than overall accuracy alone.

Collectively, this methodology supports rigorous empirical assessment of crash severity determinants while maintaining transparency, reproducibility, and alignment with applied transportation safety objectives.

The analytical workflow was implemented in Python, and the core preprocessing, feature engineering, and modeling steps are provided in Appendix C.

Chapter 4: Results

This chapter presents the empirical results of the study, beginning with descriptive statistics and progressing through model performance, predictor interpretation, robustness checks, and validation findings. The results are organized to show both the structure of the crash dataset and the comparative performance of the modeling frameworks used to examine motorcycle injury severity in Southern California. Together, these findings address the study's research questions and provide the basis for the interpretations presented in Chapter 5.

4.1 Descriptive Statistics:

Crash configuration patterns reveal that the study sample is primarily characterized by multi-vehicle collisions involving directional conflicts. Broadside collisions account for 24.9% while same-direction sideswipe incidents make up 20.7% and rear-end crashes constitute 19.0%. In reference to Table 7, These categories align directly with the dummy variables identified through correlation analysis such as TYPE_OF_COLLISION_B, TYPE_OF_COLLISION_D, and TYPE_OF_COLLISION_E. This alignment confirms that the most prevalent types of crash are also the most statistically significant for predicting injury severity. In contrast, less common configurations like overturned crashes (0.8%) and vehicle-pedestrian collisions (4.2%) rarely occur but can be detrimental to survivability.

Table 7: *Normalized Summary Statistics Categorical TYPE_OF_COLLISION*

Variable	Class	Class Meaning	Proportion of N = 19,721
TYPE_OF_COLLISION	B	Broadside	0.249
TYPE_OF_COLLISION	D	Sideswipe same direction	0.207
TYPE_OF_COLLISION	C	Rear-End	0.190
TYPE_OF_COLLISION	F	Hit Object	0.152
TYPE_OF_COLLISION	E	Sideswipe, Opposite Direction	0.101
TYPE_OF_COLLISION	A	Head-On	0.045
TYPE_OF_COLLISION	H	Vehicle/Pedestrian	0.042
TYPE_OF_COLLISION	G	Overturned	0.008
TYPE_OF_COLLISION	-	Not Stated	0.005

The PCF distribution seen in Table 8 is heavily right skewed with unsafe speed (29.4%), improper turning (20.6%), and right-of-way violations (16.2%) accounting for most hazardous behaviors. These categories represent the most common behavioral contributors to motorcycle crashes in SWITRS and correspond directly to the dummy variables included in the model, specifically PCF_VIOL_CATEGORY_01 and PCF_VIOL_CATEGORY_07. DUI incidents (4.3%) are relatively infrequent but are significantly linked to higher injury severity. The long tail of low-frequency categories emphasizes the need for careful dummy-variable selection as rare categories can introduce sparse-data bias.

Table 8: *Normalized Summary Statistics Categorical PCF_VIOL_CATEGORY (If P > 0.004)*

Variable	Class	Class Meaning	Proportion of N = 19,721
PCF_VIOL_CATEGORY	03	Unsafe Speed	0.294
PCF_VIOL_CATEGORY	08	Improper Turning	0.206
PCF_VIOL_CATEGORY	09	Auto Right-of-Way	0.162
PCF_VIOL_CATEGORY	07	Unsafe Lane Change	0.115
PCF_VIOL_CATEGORY	12	Signals/Signs	0.044
PCF_VIOL_CATEGORY	01	DUI	0.043
PCF_VIOL_CATEGORY	00	Other	0.027
PCF_VIOL_CATEGORY	06	Improper Passing	0.023
PCF_VIOL_CATEGORY	05	Wrong Side of Road	0.018
PCF_VIOL_CATEGORY	18	Other Then Driver	0.016
PCF_VIOL_CATEGORY	17	Other Hazardous Violation	0.011
PCF_VIOL_CATEGORY	04	Following Too Closely	0.009
PCF_VIOL_CATEGORY	21	Fell Asleep	0.007
PCF_VIOL_CATEGORY	22	Unknown	0.005
PCF_VIOL_CATEGORY	-	Not Stated	0.005

The CVC violation codes seen in Table 9 reflect the PCF distribution closely, with three sections - 22350 (Basic Speed Law), 22107 (Unsafe Turn), and 21658 (Unsafe Lane Change), accounting for nearly 60% of all violations. These codes correspond directly to the dummy variables selected for modeling, specifically PCF_VIOLATION_21658.0 and PCF_VIOLATION_23152.0, reinforcing their statistical and substantive relevance. The concentration of violations in a small number of high-frequency codes reflects typical urban traffic patterns and highlights the behavioral mechanisms behind many motorcycle crashes. The presence of "Unknown" violations (4.9%) indicates SWITRS reporting limitations but does not significantly impact the dominant behavioral trends.

Table 9: *Normalized Summary Statistics Categorical PCF_VIOLATION (Top 5)*

Variable	Class	Class Meaning	Proportion of N = 19,721
PCF_VIOLATION	22350.0	Basic Speed Law	0.295
PCF_VIOLATION	22107.0	Unsafe Turn	0.192
PCF_VIOLATION	21658.0	Unsafe Lane Change	0.115
PCF_VIOLATION	21801.0	Left Turn/U-Turn	0.087
PCF_VIOLATION	Unknown	Not Reported	0.049

Temporal patterns indicate that crashes are primarily concentrated during Midday (30.9%) and Evening (31.3%). The combined account for over 60% of all events. Nighttime crashes represent 18.3% of the dataset and are included in the dummy variable TIME_OF_DAY_Night, which was retained due to its correlation with injury severity. While nighttime crashes occur less frequently, they are consistently linked to higher severity in SWITRS data due to factors such as reduced visibility, fatigue, and increased rates of impaired driving.

Table 10: *Normalized Summary Statistics Categorical TIME_OF_DAY*

Variable	Class	Class Meaning	Proportion of N = 19,721
TIME_OF_DAY	Evening	4 PM – 8 PM	0.313
TIME_OF_DAY	Midday	10 AM – 4 PM	0.309
TIME_OF_DAY	Morning	5 AM -10 AM	0.194
TIME_OF_DAY	Night	8 PM – 5 AM	0.183

Alcohol involvement is coded as "Unknown" (92.5%), a known SWITRS limitation that describes MNAR (Missing-Not-At-Random) reporting rather than true absence of alcohol. Confirmed cases are coded as "Y," representing 7.5% of the sample. The dummy variable

ALCOHOL_INVOLVED_Y captures a small but meaningful subset of crashes where alcohol involvement was verified. This variable must be interpreted cautiously in modeling, as the high rate of missingness limits its representativeness.

Table 11: *Normalized Summary Statistics Categorical ALCOHOL_INVOLVED*

Variable	Class	Class Meaning	Proportion of N = 19,721
ALCOHOL_INVOLVED	Unknown	Not Reported	0.925
ALCOHOL_INVOLVED	Y	Alcohol Was Involved	0.075

Greater than half of all crashes (55.9%) required towing. This shows most of all crash events involved deacceleration forces which did not allow the motorcycle to leave the scene without assistance. This aligns with the elevated severity typically observed in motorcycle crashes and supports the inclusion of *TOW_AWAY_Y* as a severity-related predictor.

Table 12: *Normalized Summary Statistics Categorical TOW_AWAY*

Variable	Class	Class Meaning	Proportion of N = 19,721
TOW_AWAY	Y	Vehicle Towed	0.559
TOW_AWAY	N	Not Towed	0.434
TOW_AWAY	Unknown	Not Reported	0.007

Lighting conditions as seen in Table 13 reveal that 68.7% of crashes happen in daylight while 21.2% occur in dark unlit environments and 5.9% in dark street-lit conditions. The dummy variables LIGHTING_A and LIGHTING_D were retained for their statistical relevance and interpretive clarity. These lighting patterns align with SWITRS trends showing higher daytime exposure but often more severe nighttime crashes.

Table 13: *Normalized Summary Statistics Categorical LIGHTING*

Variable	Class	Class Meaning	Proportion of N = 19,721
LIGHTING	A	Daylight	0.687
LIGHTING	C	Dark, No Streetlights	0.212
LIGHTING	D	Dark, Streetlights On	0.059

LIGHTING	B	Dusk/Dawn	0.038
LIGHTING	-	Not Stated	0.002
LIGHTING	E	Dark, No Streetlights	0.002

Table 14: *Normalized Summary Statistics Categorical HIT_AND_RUN*

Variable	Class	Class Meaning	Proportion of N = 19,721
HIT_AND_RUN	N	Not Hit and Run	0.892
HIT_AND_RUN	F	Felony Hit and Run	0.086
HIT_AND_RUN	M	Misdemeanor Hit-and-Run	0.022

The city variable described in Table 15 encompasses 186 distinct geographic classes. The CITY_UNINCORPORATED dummy variable captures 19.5% of cases and reflects crashes occurring outside incorporated city limits which often vary in roadway design, enforcement, and lighting infrastructure. While large cities do contribute, the distribution shows that the dataset covers a wide range of urban and rural environments.

Table 15: *Normalized Summary Statistics Categorical CITY (P>0.020)*

Variable	Class	Class Meaning	Proportion of N = 19,721
CITY	Unincorporated	County Area	0.195
CITY	Los Angeles	Incorporated City	0.143
CITY	San Diego	Incorporated City	0.093
CITY	Long Beach	Incorporated City	0.028
CITY	Riverside	Incorporated City	0.021

Table 16 conveys that Motorcycles are coded as the at-fault vehicle in 53.5% of cases which accurately describe the motorcycle focused and lensed TIMS dataset. Passenger cars account for 30.2% of at-fault designations. The dummy variable *STWD_VEHTYPE_AT_FAULT_C* captures the most prevalent at-fault category and was retained due to its strong correlation with injury severity

Table 16: *Normalized Summary Statistics Categorical STWD_VEHTYPE_AT_FAULT*

Variable	Class	Class Meaning	Proportion of N = 19,721
STWD_VEHTYPE_AT_FAULT	C	Motorcycle	0.535
STWD_VEHTYPE_AT_FAULT	A	Passenger Car	0.302
STWD_VEHTYPE_AT_FAULT	-	Unknown	0.109
STWD_VEHTYPE_AT_FAULT	D	Truck/Tractor	0.035
STWD_VEHTYPE_AT_FAULT	O	Other	0.005
STWD_VEHTYPE_AT_FAULT	Remaining Classes	Rare Types	<0.004

The standardized TIMS field aggregates all motorcycle-related codes, revealing that motorcycles are listed as the at-fault vehicle in 53.5% of incidents. Conversely, the California Highway Patrol (CHP) field seen in Table 17 employs more detailed officer-entered codes, identifying passenger cars as the predominant at-fault vehicle type at 47.7%. According to this distribution, motorcycles (Code 01) represent 21.9% of cases, while buses (Code 07) account for 6.6%. Additionally, 5.9% of cases are marked as "Unknown," and 4.4% fall into the "Other" category. The dummy variable CHP_VEHTYPE_AT_FAULT_02 reflects the most substantial and stable at-fault category, emphasizing that passenger cars often cause motorcycle crashes due to issues like turning conflicts, lane-change mistakes, and visibility limitations. Here, motorcycles are at-fault vehicles in approximately 20% of cases. CHP is a more granular coding system, and the distribution has minimal missing data for better model fidelity. CHP coding better shows passenger car involvement as a significant factor in the injury-severity model.

Table 17: *Normalized Summary Statistics Categorical CHP_VEHTYPE_AT_FAULT*

Variable	Class	Class Meaning	Proportion of N = 19,721
CHP_VEHTYPE_AT_FAULT	02	Passenger Car	0.477
CHP_VEHTYPE_AT_FAULT	01	Motorcycle	0.219
CHP_VEHTYPE_AT_FAULT	07	Bus	0.066
CHP_VEHTYPE_AT_FAULT	-	Unknown	0.059
CHP_VEHTYPE_AT_FAULT	99	Other	0.044
CHP_VEHTYPE_AT_FAULT	Remaining Codes	Rare Types	<0.004

4.2 Model Performance Metrics

This section reports the predictive performance of the three classification models developed in this study: Logistic Regression, Random Forest, and XGBoost. Consistent with the evaluation framework established in Chapter 3, model performance was assessed using accuracy, precision, recall, F1 score, and ROC–AUC on the held-out test set, *n* = 2,958. Because the KABCO injury severity outcome remained imbalanced in its natural form, model comparison emphasized not only overall accuracy but also class-sensitive measures that better reflect performance across all four outcome categories. Macro-averaged F1 was designated as the primary comparative metric because it gives equal weight to each injury severity class, including the less frequent fatal injury class.

Table 18 presents aggregate performance metrics for all three models on the held-out test set. Logistic regression reported the lowest values across all summary metrics. Random Forest and XGBoost each improved on logistic regression in terms of accuracy, macro F1, and ROC–AUC, while remaining close to one another across most individual measures.

Table 18: *Overall test-set model performance*

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted Precision	Weighted Recall	Weighted F1	ROC–AUC (Weighted)	ROC–AUC (Macro)
Logistic Regression	0.46	0.39	0.35	0.35	0.44	0.46	0.43	0.6253	0.6505
Random Forest	0.48	0.41	0.37	0.37	0.47	0.48	0.44	0.6606	0.6941
XGBoost	0.48	0.43	0.36	0.36	0.48	0.48	0.44	0.6610	0.6926

Class-level results showed a consistent pattern across all three models. Performance was strongest for Class 3 (visible injury) and weaker for the minority and adjacent-severity classes. Across models, recall for Class 3 ranged from 0.73 (logistic regression) to 0.81 (XGBoost), the highest recall observed for any class. By contrast, recall for Class 1 (fatal injury) remained low across all three frameworks, ranging from 0.16 to 0.20, and recall for Class 4 (complaint of pain) similarly ranged from 0.18 to 0.21.

Precision values followed a similar pattern shown in Table 4.2.2 . Class 2 (severe injury) and Class 3 generally produced higher precision than Class 1. Class 1 precision remained comparatively low across all three frameworks, ranging from 0.23 to 0.26. These class-specific results indicate that model discrimination varied meaningfully across severity categories even when overall accuracy appeared similar.

Table 19: *Class-specific precision and recall by model*

Severity Class	Metric	Logistic Regression	Random Forest	XGBoost
Class 1 (Fatal Injury)	Precision	0.26	0.23	0.24
	Recall	0.16	0.20	0.20
Class 2 (Severe Injury)	Precision	0.44	0.49	0.52
	Recall	0.28	0.31	0.26
Class 3	Precision	0.49	0.50	0.49

(Visible Injury)				
	Recall	0.73	0.76	0.81
Class 4 (Complaint of Pain)	Precision	0.39	0.43	0.45
	Recall	0.21	0.20	0.18

ROC curves were generated for each model. Table 20 presents per class, micro-average, and macro-average AUC values for all three models. Class 1 (fatal injury) consistently showed the largest one-vs-rest AUC values across models, ranging from 0.719 (logistic regression) to 0.792 (Random Forest), while Class 3 (visible injury) showed the smallest per-class AUC values, ranging from 0.591 to 0.629. This inverse relationship between per-class recall and per-class AUC for Class 3 reflects the model's tendency to assign high predicted probabilities broadly to the majority class, inflating recall while compressing the AUC margin.

At the model level, Random Forest produced the highest macro ROC–AUC (0.6941), whereas XGBoost produced the highest weighted ROC–AUC (0.6610). The difference between Random Forest and XGBoost on these aggregate AUC measures was small, whereas both exceeded logistic regression by a wider margin.

Table 20: *Per-class one-vs-rest AUC by model*

	Logistic Regression	**Random Forest**	**XGBoost**
Class 1 (Fatal Injury)	0.719	0.792	0.789
Class 2 (Severe Injury)	0.650	0.689	0.682

Class 3 (Visible Injury)	0.591	0.622	0.629
Class 4 (Complaint of Pain)	0.642	0.674	0.670
Micro-Average AUC	0.743	0.762	0.765
Macro-Average AUC	0.651	0.694	0.693

Figures 7, 8, and 9 display the ROC curves for each model. The ROC curve for logistic regression remained closer to the diagonal reference line than the corresponding curves for Random Forest and XGBoost, particularly for Class 1 (fatal injury). The macro- and micro-average curves for Random Forest and XGBoost were closely aligned, consistent with the small numerical difference observed in Table 20.

Figure 7: *ROC curves for logistic regression*

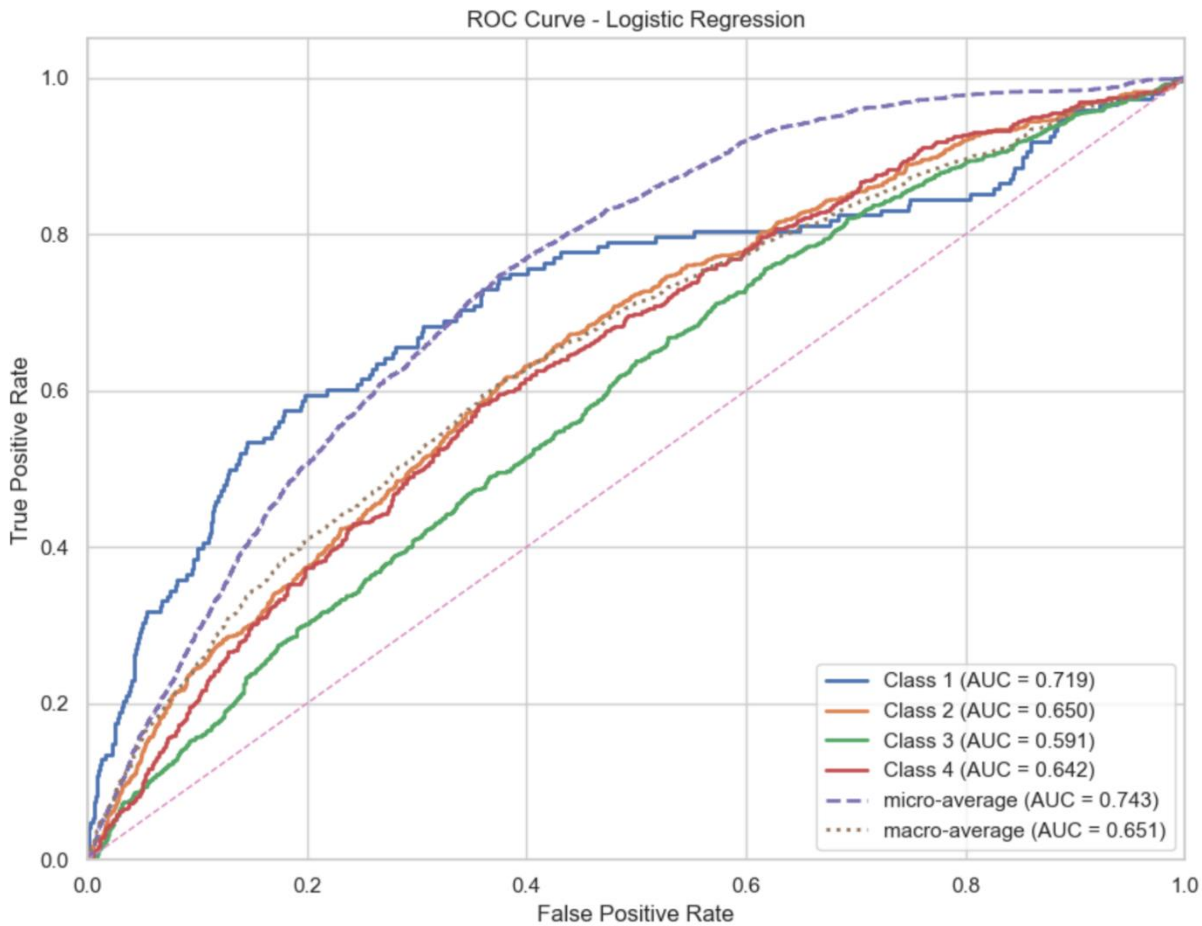


Figure 8: *ROC curves for Random Forest*

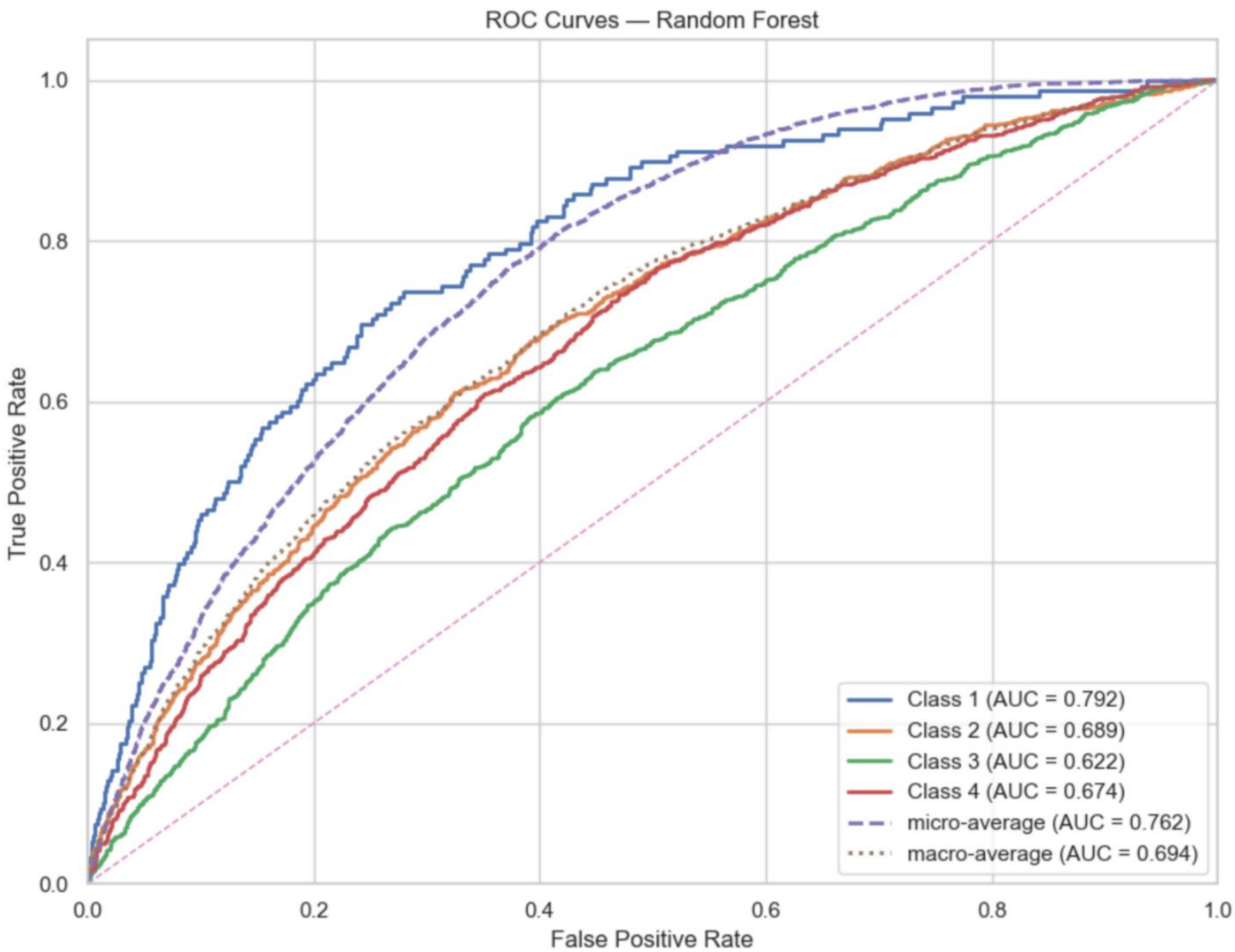
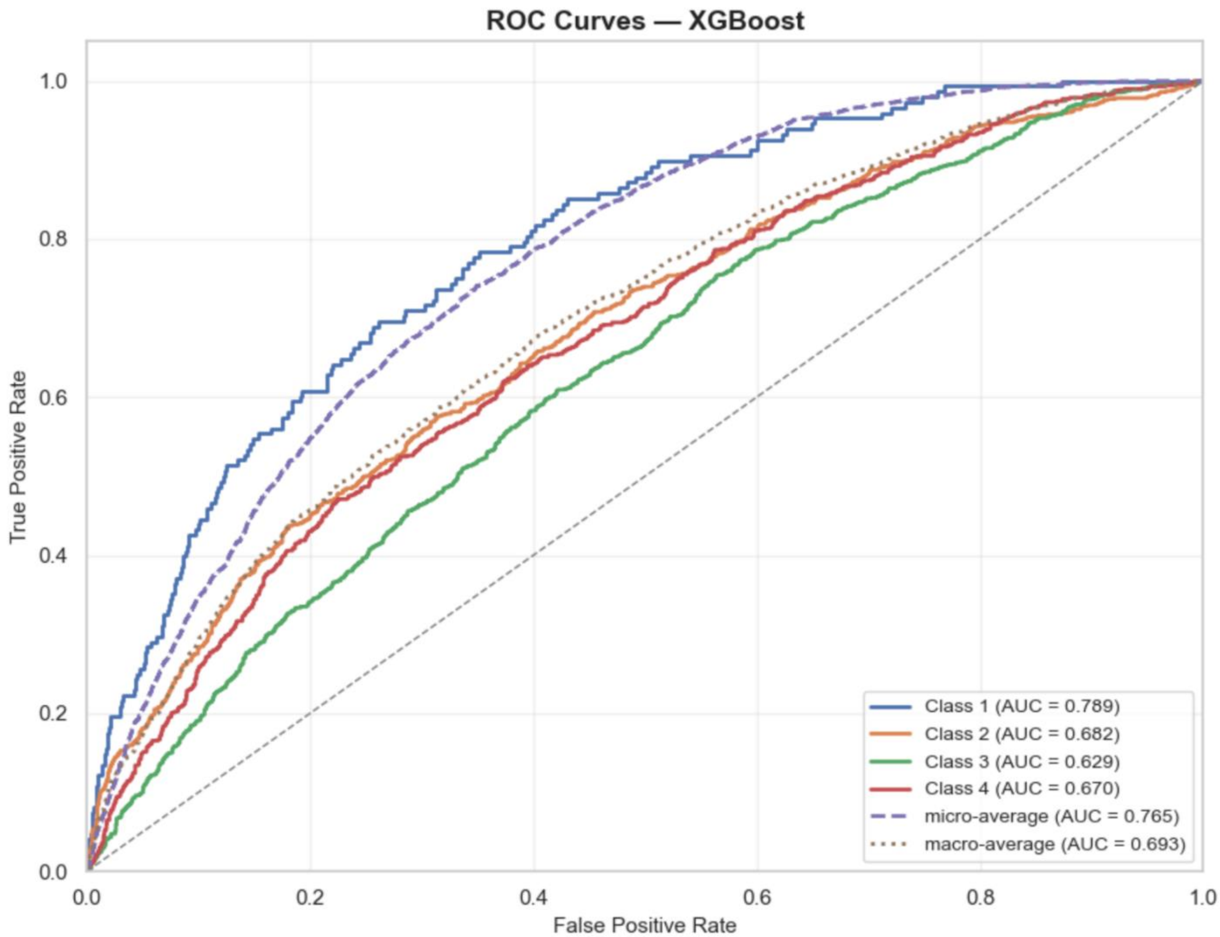


Figure 9: ROC curves for XGBoost



Taken together, the three models produced moderate predictive performance on the test set. Logistic regression reported the lowest overall discrimination and the lowest macro F1. Random Forest and XGBoost each improved on logistic regression in terms of accuracy, macro F1, and ROC–AUC, while remaining close to one another across most metrics. Random Forest reported the highest macro F1 and macro ROC–AUC, whereas XGBoost reported the highest weighted ROC–AUC. Across all models, Class 3 (visible injury) produced the highest recall, while Class 1 (fatal injury) and Class 4 (complaint of pain) remained the most difficult classes to recover at high recall levels.

4.3 Key Predictors of Injury Severity

Let's look at the variables strongly associated with injury severity in the models. This section reports rank order, direction of association, and relative contribution strength as observed in SHAP visualizations. SHAP based explanation output from the gradient-boosted tree model was examined at three levels: (1) global importance across all classes, (2) class-specific contribution patterns for Class 1 (fatal injury), and (3) a local explanation for one individual prediction.

At the global level, the SHAP ranking was dominated by a small set of variables (see Figure 4.3.1). The largest mean absolute SHAP contribution was observed for TOW_AWAY_Y (vehicle towed away), followed by TYPE_OF_COLLISION_D (broadside collision). The next highest-ranked predictors were CITY_LOS_ANGELES, TYPE_OF_COLLISION_E (hit object), TYPE_OF_COLLISION_C (rear-end), and TYPE_OF_COLLISION_B (sideswipe). Additional variables in the upper portion of the global ranking included ALCOHOL_INVOLVED_Y, CITY_UNINCORPORATED, CHP_VEHTYPE_AT_FAULT_02 (motorcycle at fault), CONTROL_DEVICE_A (no control device), LIGHTING_C (dark, no street lights), STWD_VEHTYPE_AT_FAULT_A, COLLISION_TIME, PCF_VIOLATION_22350.0 (unsafe speed), INTERSECTION_Y, LONGITUDE, and PARTY_COUNT. Table 21 lists these predictors in rank order alongside readable labels and domain classifications.

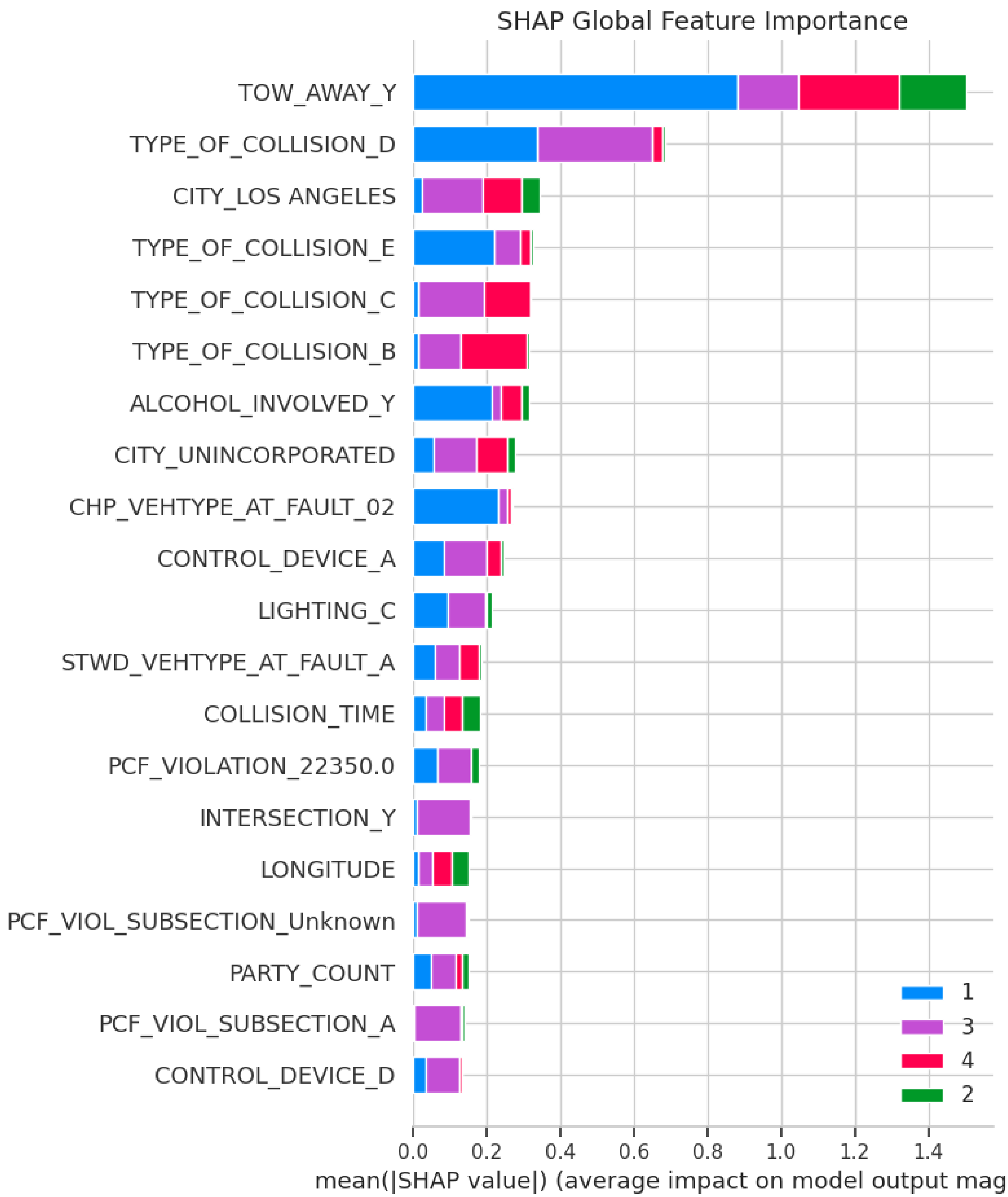
Table 21: *Top global predictors of injury severity from SHAP importance output*

Rank	Predictor (encoded)	Readable Label	Domain
1	TOW_AWAY_Y	Vehicle towed away (yes)	Crash outcome / damage

2	TYPE_OF_COLLISION_D	Broadside collision	Collision mechanism
3	CITY_LOS_ANGELES	Crash in Los Angeles	Spatial
4	TYPE_OF_COLLISION_E	Hit object collision	Collision mechanism
5	TYPE_OF_COLLISION_C	Rear-end collision	Collision mechanism
6	TYPE_OF_COLLISION_B	Sideswipe collision	Collision mechanism
7	ALCOHOL_INVOLVED_Y	Alcohol involvement (yes)	Behavioral
8	CITY_UNINCORPORATED	Unincorporated area	Spatial
9	CHP_VEHTYPE_AT_FAULT_02	Motorcycle at fault	Vehicle / roadway actor
10	CONTROL_DEVICE_A	No traffic control device	Roadway control
11	LIGHTING_C	Dark - no street lights	Environmental
12	STWD_VEHTYPE_AT_FAULT_A	Passenger car at fault	Vehicle / roadway actor
13	COLLISION_TIME	Time of crash (continuous)	Temporal
14	PCF_VIOLATION_22350.0	Unsafe speed violation	Behavioral / violation

15	INTERSECTION_Y	Crash at intersection (yes)	Roadway
16	LONGITUDE	Crash longitude coordinate	Spatial
17	PARTY_COUNT	Number of parties involved	Crash configuration

Figure 10: *Global SHAP feature importance across injury severity classes*



The global ranking showed that collision mechanism variables were especially prominent. Four collision-type indicators (TYPE_OF_COLLISION_B through TYPE_OF_COLLISION_E) appeared within the top six positions, indicating that encoded crash configuration variables accounted for a substantial share of model variation. Spatial indicators also appeared early in the ranking, with CITY_LOS_ANGELES (rank 3) and CITY_UNINCORPORATED (rank 8) both placed near the top. Behavioral and roadway-control indicators were present as well, though generally below the highest-ranked collision variables.

The Class 1 (fatal injury) SHAP summary plot provided a more detailed view of contribution patterns for the fatal-injury outcome (see Figure 10). The most prominent features in this class-specific output were TOW_AWAY_Y, TYPE_OF_COLLISION_D, CHP_VEHTYPE_AT_FAULT_02, TYPE_OF_COLLISION_E, ALCOHOL_INVOLVED_Y, LIGHTING_C, CONTROL_DEVICE_A, PCF_VIOLATION_22350.0, STWD_VEHTYPE_AT_FAULT_A, CITY_UNINCORPORATED, and PARTY_COUNT.

Additional variables appearing within the top twenty for Class 1 included PCF_VIOL_CATEGORY_06, PCF_VIOLATION_21658.0, PCF_VIOL_CATEGORY_03, PCF_VIOL_CATEGORY_01, LIGHTING_D, COLLISION_TIME, CONTROL_DEVICE_D, PCF_VIOL_CATEGORY_05, and Other Registration.

Several predictors were concentrated on the positive SHAP side in the Class 1 plot, indicating that their higher or present values were associated with an increase in the model output for Class 1. The clearest positive-shift variables were TOW_AWAY_Y, TYPE_OF_COLLISION_E, TYPE_OF_COLLISION_D, CHP_VEHTYPE_AT_FAULT_02, and ALCOHOL_INVOLVED_Y. These variables also exhibited some of the widest horizontal spreads, indicating comparatively large per-observation contribution ranges. Additional variables with predominantly positive placement included LIGHTING_C, CONTROL_DEVICE_A, CITY_UNINCORPORATED, PCF_VIOL_CATEGORY_01, PCF_VIOL_CATEGORY_03, PCF_VIOL_CATEGORY_05, and LIGHTING_D.

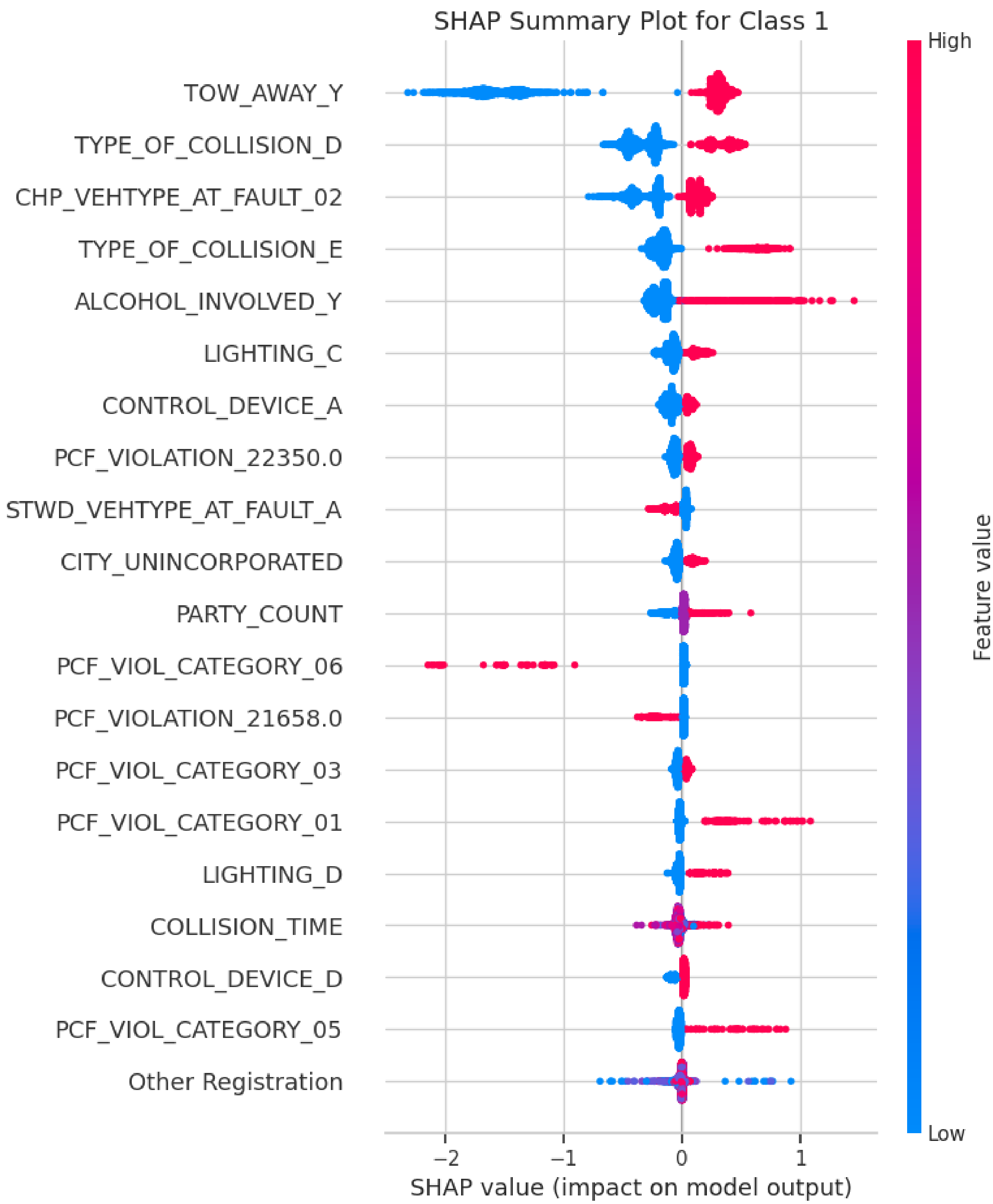
By contrast, several variables were concentrated on the negative SHAP side in the Class 1 plot. The most clearly negative patterns were observed for PCF_VIOL_CATEGORY_06 (right-of-way violation category), PCF_VIOLATION_21658.0 (unsafe lane change), and STWD_VEHTYPE_AT_FAULT_A (passenger car at fault), where higher or present values were located predominantly to the left of zero. PARTY_COUNT and COLLISION_TIME showed more mixed placement, with contributions spanning both the negative and positive sides of the class-specific output. Table 22 summarizes the dominant SHAP direction and relative strength for selected Class 1 predictors.

Table 22: *Dominant Class 1 SHAP directions for selected predictors*

Predictor (encoded)	Readable Label	Direction	Relative Strength
TOW_AWAY_Y	Vehicle towed away (yes)	Positive	Very high
TYPE_OF_COLLISION_E	Hit object collision	Positive	High
TYPE_OF_COLLISION_D	Broadside collision	Positive	High
CHP_VEHTYPE_AT_FAULT_02	Motorcycle at fault	Positive	High
ALCOHOL_INVOLVED_Y	Alcohol involvement (yes)	Positive	High

LIGHTING_C	Dark - no street lights	Positive	Moderate
CONTROL_DEVICE_A	No traffic control device	Positive	Moderate
CITY_UNINCORPORATED	Unincorporated area	Positive	Moderate
PCF_VIOL_CATEGORY_01	DUI violation category	Positive	Moderate
PCF_VIOL_CATEGORY_05	Speed violation category	Positive	Moderate
STWD_VEHTYPE_AT_FAULT_A	Passenger car at fault	Negative	Moderate
PCF_VIOL_CATEGORY_06	Right-of-way violation category	Negative	High
PCF_VIOLATION_21658.0	Unsafe lane change violation	Negative	Moderate
PARTY_COUNT	Number of parties involved	Mixed (slight positive)	Moderate
COLLISION_TIME	Time of crash (continuous)	Mixed	Low–moderate

Figure 11: *SHAP summary plot for Class 1 (fatal injury)*



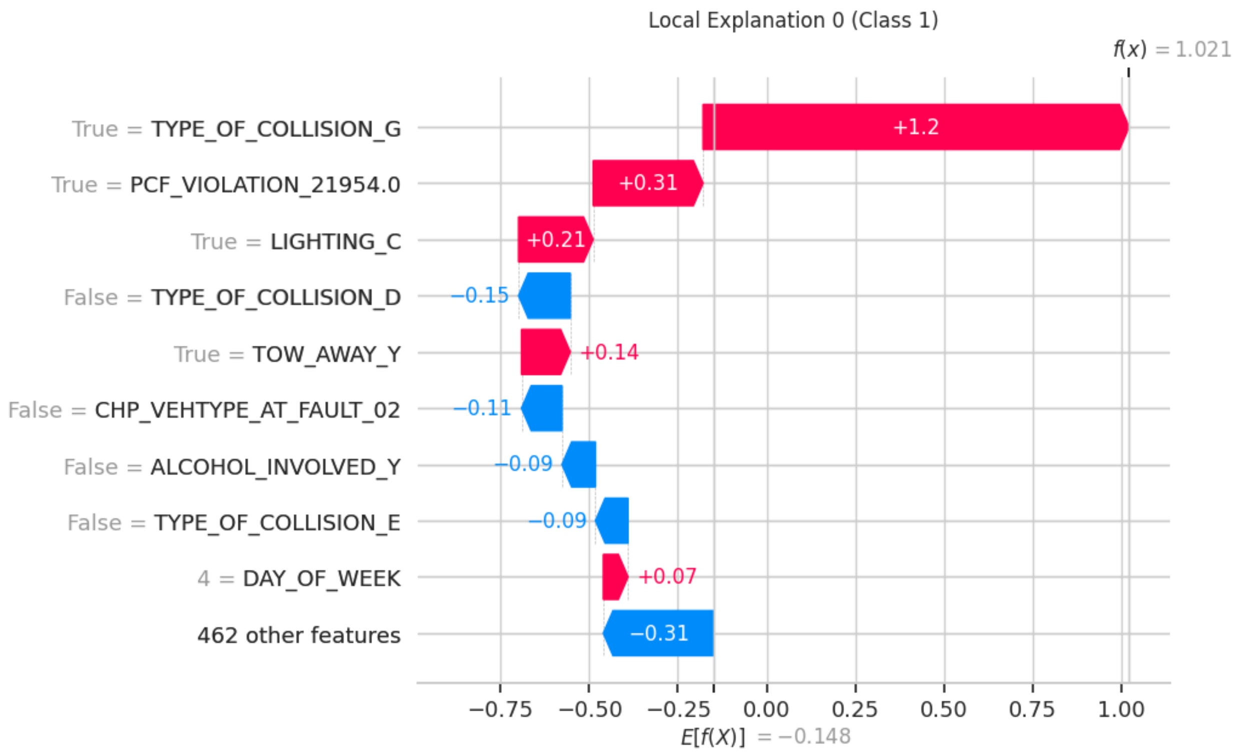
The local explanation plot for one representative Class 1 prediction illustrated how these variables combined at the individual-record level (see Table 23). In that observation, the largest positive contributor was TYPE_OF_COLLISION_G (vehicle overturned), with a SHAP contribution of approximately +1.20. Other positive contributors included PCF_VIOLATION_21954.0 (reckless driving, +0.31), LIGHTING_C (+0.21), TOW_AWAY_Y (+0.14), and DAY_OF_WEEK coded as 4-Thursday-(+0.07). On the negative side, the absence of TYPE_OF_COLLISION_D contributed approximately −0.15, the absence of CHP_VEHTYPE_AT_FAULT_02 contributed −0.11, the absence of ALCOHOL_INVOLVED_Y contributed −0.09, and the absence of TYPE_OF_COLLISION_E contributed −0.09. The aggregate effect of the remaining 462 other features was approximately −0.31. Table 23 presents these local contributors in direction order.

Table 23: *Top local contributors in one Class 1 prediction*

Direction	Predictor (encoded)	Readable Label	Approx. SHAP Value
Positive	TYPE_OF_COLLISION_G	Vehicle overturned	+1.20
Positive	PCF_VIOLATION_21954.0	Reckless driving violation	+0.31
Positive	LIGHTING_C	Dark - no street lights	+0.21
Positive	TOW_AWAY_Y	Vehicle towed away (yes)	+0.14
Positive	DAY_OF_WEEK = 4	Thursday	+0.07

Negative	not TYPE_OF_COLLISION_D	Broadside collision absent	−0.15
Negative	not CHP_VEHTYPE_AT_FAULT_02	Motorcycle at fault absent	−0.11
Negative	not ALCOHOL_INVOLVED_Y	Alcohol involvement absent	−0.09
Negative	not TYPE_OF_COLLISION_E	Hit object collision absent	−0.09
Negative	462 other features (combined)	Aggregate background effect	−0.31

Figure 12: *Local SHAP waterfall plot for one Class 1 prediction*



Across the global SHAP ranking, the Class 1 summary, and the local explanation plot, a recurring predictor set emerged. Variables that appeared consistently across all three explanation levels were: TOW_AWAY_Y, TYPE_OF_COLLISION_D, TYPE_OF_COLLISION_E, ALCOHOL_INVOLVED_Y, CHP_VEHTYPE_AT_FAULT_02, LIGHTING_C, CONTROL_DEVICE_A, CITY_UNINCORPORATED, COLLISION_TIME, and PARTY_COUNT. These variables represent the most stable predictor group in the explanation outputs and were prominent regardless of the level of analysis at which they were examined.

The strongest model-associated predictors were concentrated in five domains: collision configuration (collision-type indicators), vehicle damage and tow-away status (TOW_AWAY_Y), behavioral indicators (alcohol involvement, PCF violation categories), environmental and roadway-control variables (lighting condition, traffic control device), and spatial indicators (city and unincorporated area designations). Variables from these domains occupied the highest SHAP ranks and showed the largest contribution magnitudes in both the global and class-specific explanation plots.

4.4 Model Interpretation and Comparison

Let's compares how the three modeling frameworks, logistic regression, Random Forest, and XGBoost, represented injury severity patterns in the Southern California motorcycle crash data. Rather than identifying a single "best" model, the focus here is on how each model interprets predictors, where they show convergent patterns, and where they show divergent patterns. This is important since the three models do not express variable relationships in the same way, logistic regression yields directional class-specific estimates while Random Forest summarizes signal through ensemble tree structure and variable importance ranking and XGBoost, paired with SHAP output, provides observation level and global contribution patterns showing magnitude and direction of influence.

Predictors are consistent across models, they differ in structure but the results indicate that they show consistency most clearly at predictors. The SHAP output shows that the most significant features are concentrated in collision configuration, tow-away and damage severity, behavioral indicators, lighting and traffic-control conditions, and spatial context. Table 24 summarizes domain-level representation across all three models.

Table 24: *Predictor domain consistency across modeling frameworks*

Predictor Domain	**Example Predictors**	**Logistic Regression**	**Random Forest**	**XGBoost (SHAP)**

Collision configuration	TYPE_OF_COLLISION_B/C/D/E	Retained	Top-ranked	Top-ranked globally
Tow-away / damage severity	TOW_AWAY_Y	Retained	High importance	#1 Global SHAP rank
Behavioral / violation	ALCOHOL_INVOLVED_Y, PCF_VIOLATION_22350.0, PCF_VIOL_CATEGORY	Retained	Present	Prominent in Class 1
Environmental / roadway control	LIGHTING_C, CONTROL_DEVICE_A	Retained	Present	Positive Class 1 contributors
Spatial context	CITY_LOS_ANGELES, CITY_UNINCORPORATED, LONGITUDE	Retained	Present	Global and Class 1 ranked
Vehicle / roadway actor	CHP_VEHTYPE_AT_FAULT_02, STWD_VEHTYPE_AT_FAULT_A	Retained	Present	Prominent in Class 1
Temporal	COLLISION_TIME, DAY_OF_WEEK	Retained	Present	Mixed

The repeated importance of collision-type is noteworthy. Four collision type variables appeared within the top six positions of the global SHAP ranking, indicating that crash mechanism contributed to model variation. Second recurring cluster involved behavioral and regulatory indicators, especially alcohol involvement and violation codes. In the Class 1 SHAP summary for fatal injury, ALCOHOL_INVOLVED_Y, PCF_VIOLATION_22350.0, and several other PCF violation categories were significant contributors. Third cluster involved environmental and roadway control variables, including lighting condition and traffic-control device status. The fatal-injury SHAP output showed LIGHTING_C and CONTROL_DEVICE_A as repeated positive contributors. The models differed in how they showed that information, but the strongest signal involved crash mechanism, tow-away status, behavioral violations, lighting and control conditions, and spatial context.

Even though broad predictor domains were stable, the three models behaved differently in how they separated injury classes. As shown in Table 25, the class-level pattern was similar across frameworks, all three models performed best on Class 3 (visible injury) and struggled more with Class 1 (fatal injury) and Class 4 (complaint of pain). Recall for Class 3 ranged from 0.73 in logistic regression to 0.81 in XGBoost, whereas recall for Class 1 ranged only from 0.16 to 0.20. This shared pattern indicates that the models were detecting a common structure in the data.

Table 25: *Performance metric comparison across models*

Metric	Logistic Regression	Random Forest	XGBoost
Accuracy	0.46	0.48	0.48
Macro Precision	0.39	0.41	0.43

Macro Recall	0.35	0.37	0.36
Macro F1	0.35	0.37	0.36
Weighted F1	0.43	0.44	0.44
ROC-AUC (Macro)	0.6505	0.6941	0.6926
ROC-AUC (Weighted)	0.6253	0.6606	0.6610
Class 1 Recall	0.16	0.20	0.20
Class 3 Recall	0.73	0.76	0.81

At the same time, the models differs where logistic regression produced the lowest overall discrimination, accuracy = 0.46, macro F1 = 0.35, macro ROC-AUC = 0.6505, and its ROC curves remained closer to the diagonal reference line than the corresponding curves for Random Forest and XGBoost. This pattern is consistent with a model that shows more relationship between predictors but less flexible in capturing interactions. Random Forest and XGBoost behaved similarly to one another. Random Forest showed macro F1 = 0.37 and macro ROC-AUC = 0.6941, while XGBoost showed macro F1 = 0.36 and macro ROC-AUC = 0.6926. Their ROC curves were closely aligned, suggesting that both tree-based methods captured comparable structure in the data through different algorithms in Random Forest and XGBoost.

The difference between the two methods shows in how they distribute across classes. Random Forest showed the highest macro ROC-AUC, while XGBoost showed the highest weighted ROC-AUC. Because macro metrics give equal emphasis to each class and weighted metrics reflect class prevalence, this pattern suggests that Random Forest provided slightly more balanced discrimination across classes, whereas XGBoost placed somewhat more effective probability ordering on the overall class distribution. This is a difference in model behavior, not in substantive conclusion: both models identified broadly similar severity structure, but emphasized it somewhat differently in aggregate evaluation.

The most important comparison across models is the tradeoff between interpretability and predictive flexibility. Logistic regression is the most transparent model because it supports direct interpretation and statistical inference, making it especially useful when the research aim includes identifying whether predictor effects are positive or negative for a given outcome category. The cost of that transparency is reduced flexibility: logistic regression cannot naturally represent complex interaction surfaces unless those terms are manually specified.

Random Forest occupies a middle position in this tradeoff. Relative to logistic regression, it is less transparent because it does not produce a single coefficient table with signed linear effects. However, it remains more interpretable than many other machine-learning approaches because it provides stable feature ranking, class behavior summaries, and discrimination patterns. In this study, Random Forest improved on logistic regression in both macro F1 and ROC-AUC while preserving a understandable interpretation structure.

XGBoost offered the most detailed model based explanation output in the current analysis because SHAP visualizations were generated for the boosted-tree results. Those figures provided three forms of interpretation not available from the ROC summaries alone: global feature importance, class-specific contribution direction, and local case-level decomposition. For example, TOW_AWAY_Y, collision-type variables, alcohol involvement, lighting, and violation indicators could be examined not only in terms of rank, but also in terms of whether their presence shifted model output positively or negatively for the fatal-injury class. This makes the boosted tree model less transparent at the raw algorithmic level, but highly informative once paired with SHAP-based explanation tools.

The tradeoff in this study was not a simple contrast between "interpretable" and "black box" models. Instead, the results show three distinct interpretation modes, summarized in Table 26.

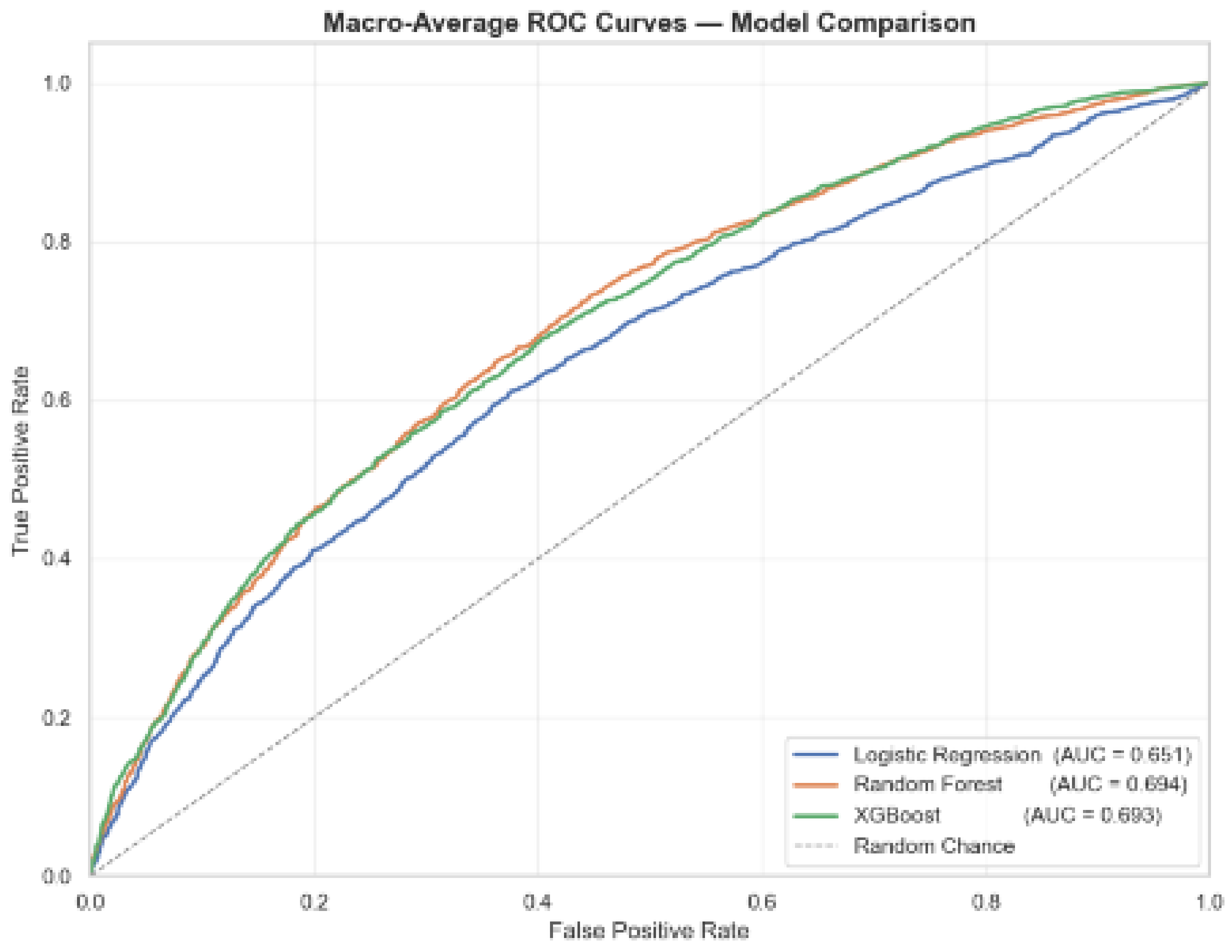
Table 26: *Interpretation modes, advantages, and limitations by modeling framework*

Model	Primary Interpretation Mode	Main Advantage	Main Limitation
Logistic Regression	Coefficients and class-specific log-odds	Highest structural transparency; supports signed directional inference	Limited nonlinear and interaction capture without manual specification
Random Forest	Ensemble-based variable importance and class behavior	Flexible nonlinear detection with moderate interpretability and stable importance ranking	No direct signed coefficient interpretation
XGBoost	SHAP-based global, class-specific, and local contributions	Richest contribution-level explanation in nonlinear space; reveals direction and magnitude	Lower raw algorithmic transparency without post hoc explanation tools

This pattern is consistent with the analytical roles, logistic regression for baseline interpretable estimation, Random Forest for nonlinear structure detection, and XGBoost for predictive optimization with SHAP explanation.

The ROC plot show that logistic regression had flatter class curves overall, while Random Forest and XGBoost demonstrated stronger separation from the diagonal, especially for Class 1 (fatal injury). This visual difference reinforces the numerical finding that tree-based models captured additional class-separation structure beyond the parametric baseline. Figure 13 places the three macro-average ROC curves side by side for direct comparison.

Figure 13: *Macro-average ROC curves for logistic regression, Random Forest, and XGBoost*



The SHAP figures add a second visual layer by showing which predictors carried the class-separation signal. The global importance plot concentrated on TOW_AWAY_Y and multiple collision-type indicators, while the Class 1 summary plot showed that the fatal-injury signal was shaped most strongly by tow-away status, collision mechanism, at-fault vehicle type, alcohol involvement, lighting, and selected violation categories. The local plot then showed how those factors combined for one individual fatal-injury prediction. Taken together, these figures demonstrate that ROC curves describe *how well* models discriminate, whereas SHAP plots describe *what* the boosted-tree explanation output relied on in producing that discrimination.

Logistic regression provided the clearest parametric interpretation but the weakest discrimination. Random Forest and XGBoost produced similar discrimination profiles, with Random Forest showing slightly stronger macro-level balance and XGBoost showing slightly stronger weighted discrimination. The boosted-tree explanation output added the most detailed contribution view of predictors, especially for fatal injury.

4.5 Sensitivity and Robustness Checks

This section looks at whether the findings of the study is stable under different modeling conditions, class-imbalance handling choices, and evaluation. Robustness is assessed by examining whether the main results remained consistent across alternative model specifications, across different performance metrics, and across the visual patterns observed in the confusion matrices. A key point is that the conclusion did not depend on a single algorithm. Across logistic regression, Random Forest, and XGBoost, the same general structure was observed where performance was strongest for Class 3 (visible injury), lower for Class 2 (severe injury), and weakest for Class 1 (fatal injury) and Class 4 (complaint of pain). This pattern appeared consistently in precision, recall, and ROC-AUC, and remained visible when outputs were examined through confusion matrices.

Robustness was first evaluated across the three models, despite structural differences the overall performance profile was stable. Logistic regression reported the weakest overall discrimination across all metrics, while Random Forest and XGBoost both produced modest improvements in accuracy, macro F1, and ROC-AUC. Critically, the substantive ordering of injury classes remained unchanged across all three models. Stability is important because it implies that the main severity pattern was not an product of one specific algorithm rather all three models continued to recover Class 3 most effectively and continued to struggle most with Classes 1 and 4. Robustness was also evaluated with respect to the study's class-imbalance handling strategy. The original training set was highly imbalanced, with class counts of 663 (fatal injury), 3,496 (severe injury), 6,189 (visible injury), and 3,456 (complaint of pain). After applying SMOTE to the training partition, each class was balanced to 6,189 observations as shown in Figure 4.5.1. Importantly, the held-out validation and test sets were left in their natural distributions, preserving realistic class proportions during model evaluation.

Figure 14: *Before and After SMOTE*

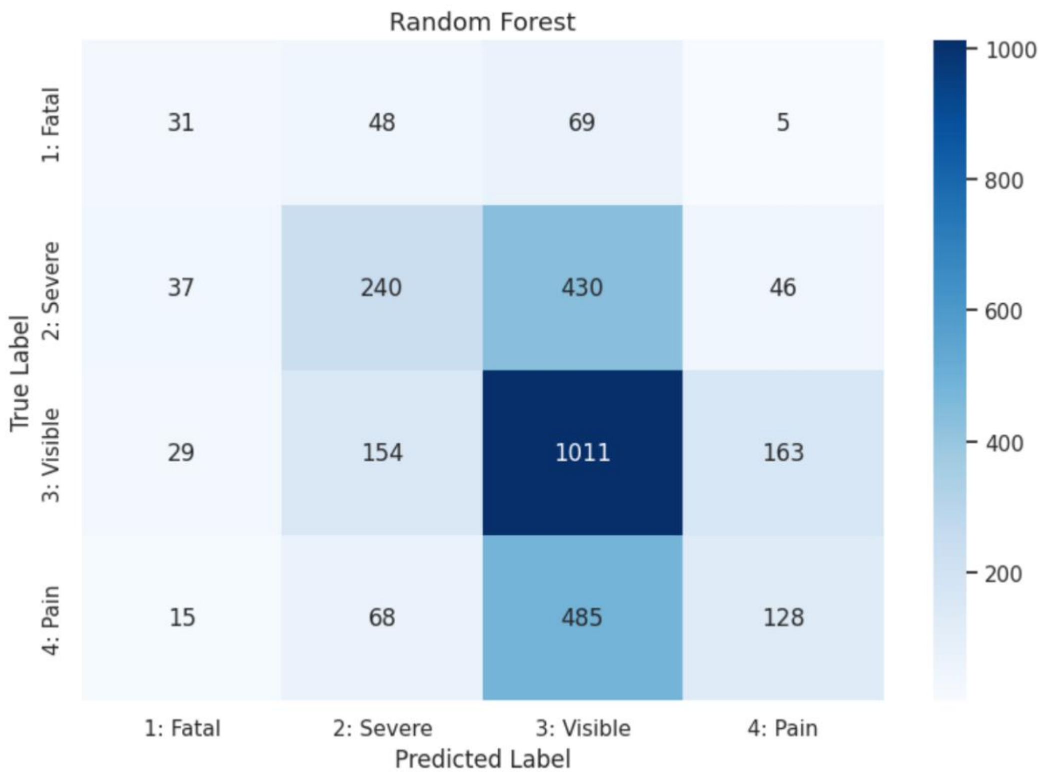
```
Original Training Set:
COLLISION_SEVERITY
1      663
2     3496
3     6189
4     3456
```

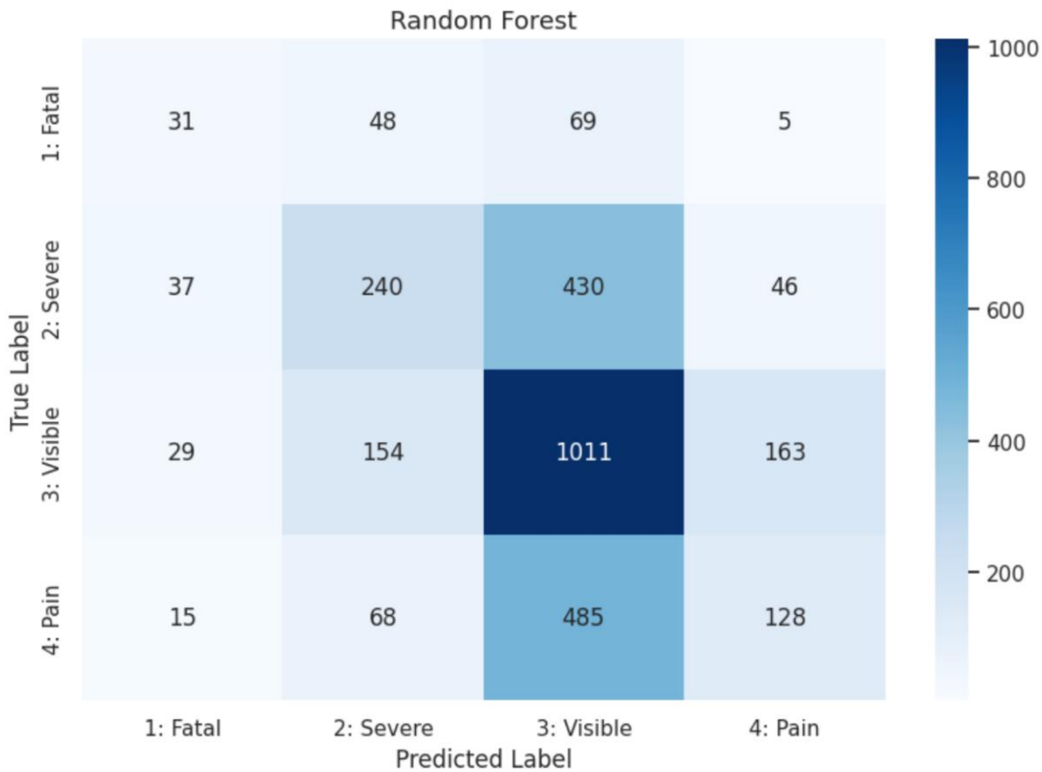
```
After SMOTE Training Set:
COLLISION_SEVERITY
1     6189
2     6189
3     6189
4     6189
```

The SMOTE resampling procedure changed the training but not the evaluation, therefore test set performance still reflected model behavior under the original population class imbalance. Under this setup, the findings remained stable where visible injury continued to dominate model recall, fatal injury remained difficult to recover, and complaint-of-pain cases were frequently put into the visible-injury category. The main conclusions were not dependent on evaluating the models on an artificially balanced test set but they were more demanding and realistic class distribution used for out of sample assessment.

The confusion matrices provide an additional robustness check by showing whether the error structure remained consistent across models. Figure 4.5.2 displays the three confusion matrices side by side. In all three, the most common prediction was Class 3 (visible injury) not only for true Class 3 cases but also for fatal, severe, and complaint-of-pain cases that were misclassified. The dominant off-diagonal error route was therefore toward Class 3 across all three modeling frameworks.

Figure 15: *Confusion matrices*




Random Forest
True Label
1: Fatal
2: Severe
3: Visible
4: Pain
31 48 69 5
37 240 430 46
29 154 1011 163
15 68 485 128
1: Fatal
2: Severe
3: Visible
4: Pain
Predicted Label
1000
800
600
400
200

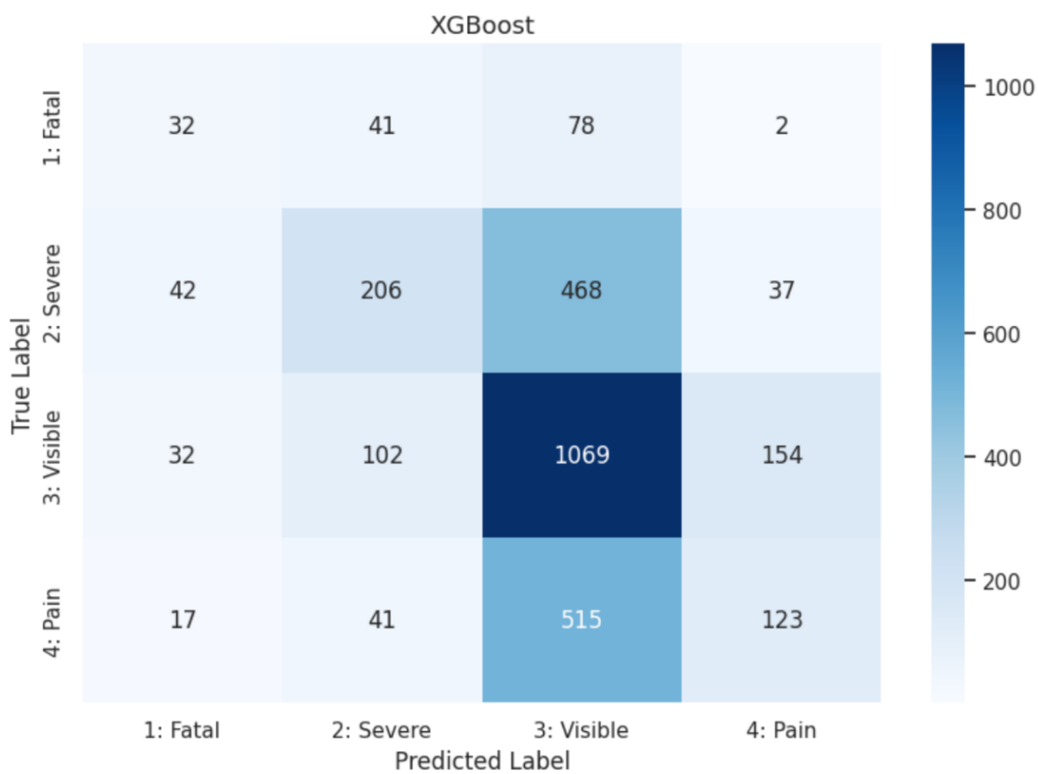

XGBoost
True Label
1: Fatal
2: Severe
3: Visible
4: Pain
32 41 78 2
42 206 468 37
32 102 1069 154
17 41 515 123
1: Fatal
2: Severe
3: Visible
4: Pain
Predicted Label
1000
800
600
400
200

Table 27 reports the specific case counts for the most common misclassification in each true class. The pattern was consistent where fatal, severe, and pain cases were most often assigned to visible injury, while true visible-injury cases most often spilled into the complaint-of-pain category. Although the exact counts varied across models reflected in recall and precision the direction and structure of errors did not change. Logistic regression, Random Forest, and XGBoost shared the same broad error, over-prediction of visible injury and adjacent-category spillover between Classes 3 and 4.

Table 27: *Dominant off-diagonal error counts*

Misclassification (True → Most Common Predicted)	**Logistic Regression**	**Random Forest**	**XGBoost**
Class 1 (Fatal) → predicted Class 3 (Visible)	65	69	78
Class 2 (Severe) → predicted Class 3 (Visible)	411	430	468
Class 3 (Visible) → predicted Class 4 (Pain)	192	163	154
Class 4 (Pain) → predicted Class 3 (Visible)	428	485	515

The robustness checks also clarify the limits of that stability. The tree-based models improved discrimination modestly, but they did not fundamentally alter the dominant error pattern. Logistic regression, Random Forest, and XGBoost differed in strength of discrimination but are on the same broad severity structure. The visual outputs reinforced this stability by showing nearly identical misclassification pathways across models.

4.6 Model Training and Validation Results

This section presents the outcomes of the model training and validation procedures. Results are reported for all three models, logistic regression, Random Forest, and XGBoost. The focus is on how the models performed during training and validation, how those results translated to the final held-out test partition, and whether the observed patterns provide evidence of model stability or overfitting.

The partitioning process yielded 13,804 observations in the initial training set, 2,958 observations in the validation set, and 2,959 observations in the held-out test set. Before resampling, the training partition was imbalanced, with 663 fatal injuries, 3,496 severe injuries, 6,189 visible injuries, and 3,456 complaint-of-pain cases. After SMOTE was applied to the training partition only, each class was balanced to 6,189 observations, yielding a resampled training set of 24,756 observations. The validation and test partitions retained the original distribution and were not resampled.

During model development, 5-fold cross-validation was conducted on the resampled training data. Figure 4.6.1 presents the per-fold scores, mean macro F1, and standard deviation for each model. Fold-to-fold variation was small for all three models, with standard deviations ranging from 0.003 to 0.005, indicating stable performance across training folds under the resampled class structure. Random Forest produced the highest mean cross-validated macro F1 (0.613), followed closely by XGBoost (0.609). Logistic regression reported a substantially lower mean (0.275).

Figure 16: *Cross-validation for all three models*

```
Logistic Regression
  Fold Scores: [0.2784 0.271  0.275  0.2723 0.2791]
  Mean: 0.2752 | Std Dev: 0.0032
```

```
Random Forest
  Fold Scores: [0.6092 0.6086 0.6122 0.6208 0.6135]
  Mean: 0.6129 | Std Dev: 0.0044


XGBoost
  Fold Scores: [0.6047 0.6129 0.6128 0.6013 0.6148]
  Mean: 0.6093 | Std Dev: 0.0053
```

After training on the resampled training partition, the models were evaluated on the natural, unaltered validation set. Table 28 summarizes the principal validation metrics for Class 1 (fatal injury) and Class 3 (visible injury).

Table 28: *Validation-set performance results*

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Class 1 Recall	Class 3 Recall
Logistic Regression	0.46	0.39	0.35	0.35	0.43	0.16	0.73
Random Forest	0.48	0.42	0.37	0.37	0.45	0.20	0.77
XGBoost	0.48	0.43	0.37	0.36	0.44	0.22	0.81

On the validation, logistic regression achieved accuracy = 0.46 and macro F1 = 0.35. Random Forest improved to accuracy = 0.48 and macro F1 = 0.37, while XGBoost reported accuracy = 0.48 and macro F1 = 0.36. The class-level pattern was already apparent at this stage, recall for Class 3 reached 0.73, 0.77, and 0.81 for logistic regression, Random Forest, and XGBoost respectively, while recall for Class 1 remained low at 0.16, 0.20, and 0.22.

Table 29 compares validation and held-out test accuracy and macro F1 directly. The transition from validation to test performance was highly stable across all three models. The absolute change in accuracy ranged from −0.003 for Random Forest to +0.029 for logistic regression and the absolute change in macro F1 ranged from −0.004 for Random Forest to +0.027 for logistic regression. None of these differences indicate a meaningful departure from validation performance.

Table 29: *Validation-versus-test performance comparison*

Model	**Validation Accuracy**	**Test Accuracy**	**Accuracy**	**Validation Macro F1**	**Test Macro F1**	**Macro F1**
Logistic Regression	0.460	0.489	+0.029	0.350	0.377	+0.027
Random Forest	0.480	0.477	−0.003	0.370	0.366	−0.004
XGBoost	0.480	0.483	+0.003	0.360	0.362	+0.002

Taken together, the results show limited evidence of overfitting in the final models. Random Forest and XGBoost, mean 5-fold cross-validation macro F1 values on the resampled training data (0.613 and 0.609, respectively) were considerably higher than their held-out validation and test macro F1 values due to SMOTE balanced training partition, where each class contributed equally, whereas the validation and test sets preserved the original imbalanced class distribution. The cross-validation therefore reflect performance under a more balanced learning environment and are not directly comparable to the held-out results as estimates of the same target distribution. The most defensible evidence regarding generalization therefore comes from the validation-to-test comparison, not from the absolute gap between cross-validation and out-of-sample scores.

The model training and validation process yielded three principal findings. First, Random Forest and XGBoost consistently outperformed logistic regression during training phase cross-validation and on the held-out validation. Second, the transition from validation to final test performance was highly stable across all three models, indicating that the selected models generalized reasonably well to unseen data. Third, the same class-specific pattern appeared throughout model development where all three models most effectively recovered Class 3 (visible injury), while Class 1 (fatal injury) remained the most difficult class to identify at high recall.

Chapter 5: Conclusions and Recommendations

This chapter synthesizes the major findings of the study, interprets them in relation to prior research, and outlines the practical and methodological implications that follow from the results. It also identifies the study's limitations and presents recommendations for future research on motorcycle injury severity in Southern California and related contexts.

5.1 Summary of Findings

This study examined motorcycle injury severity in Southern California using police-reported crash records from TIMS/SWITRS and county-level motorcycle registration data from the California DMV. Injury severity was modeled as a four-category KABCO outcome, and the analysis combined multinomial logistic regression, Random Forest, and XGBoost to evaluate both explanatory relationships and predictive performance. Across the full analytical dataset of 19,721 motorcycle crash records, the results showed that injury severity was not driven by a single condition, but instead reflected the combined influence of collision geometry, behavioral risk factors, environmental conditions, spatial context, and crash intensity.

The descriptive results established the broader crash context in which the models operated. Broadside collisions accounted for the largest share of crash configurations at 24.9%, unsafe speed was the most common primary collision factor at 29.4%, nighttime crashes represented 18.3% of observations, confirmed alcohol involvement was recorded in 7.5% of cases, and more than half of crashes involved a tow-away outcome. These patterns showed that the study area was characterized by a mix of common directional conflict crashes, frequent speed-related violations, and a substantial share of crashes involving conditions already associated with elevated injury severity in prior literature.

The modeling results reinforced these descriptive patterns. Across the three modeling approaches, a consistent set of predictors emerged as most important. Broadside collision geometry stood out as one of the strongest contributors to severe outcomes, not because it had the highest fatality proportion of any single crash type, but because it combined both high frequency and elevated severity. Alcohol involvement also consistently contributed to fatal injury prediction across models, despite clear limitations in the underlying reporting structure. Lighting conditions, particularly dark environments without street lighting, were associated with increased injury severity, while speed-related violations remained both common in the dataset and positively associated with more severe outcomes. Geographic variables, including Los Angeles and unincorporated areas, also ranked among the more influential predictors, suggesting that injury severity risk was not evenly distributed across the region.

The SHAP-based interpretation results helped tie these findings together by showing that the most influential predictors were not limited to one conceptual domain. Instead, the strongest contributors to injury severity spanned crash type, alcohol involvement, lighting, speed-related behavior, and geography. The tow-away indicator emerged as the strongest single predictor in the SHAP analysis. This finding required careful interpretation because tow-away status is not a pre-crash risk factor, but rather a post-crash marker of crash intensity. Its prominence therefore did not suggest causation. Instead, it reinforced a broader conclusion that higher-energy crashes are more likely to result in severe or fatal injury outcomes.

From a predictive standpoint, Random Forest and XGBoost outperformed multinomial logistic regression, with macro ROC-AUC values of 0.6941 and 0.6926, respectively, indicating moderate classification performance. At the same time, all three models struggled with the minority fatal injury class, with recall values ranging from 0.16 to 0.20 and a repeated tendency to misclassify fatal and severe injuries as visible injury. This pattern suggests that the available predictors were useful for identifying broad severity structure, but less effective at sharply separating neighboring injury classes. Even so, validation and test results remained stable across models, indicating that the selected modeling framework generalized reasonably well to unseen data and did not show major evidence of performance collapse outside the training process.

Taken together, the findings answer the study's core research questions by showing that severe motorcycle injury outcomes in Southern California are associated with observable behavioral, environmental, and roadway-related factors, and that these relationships remain meaningful across multiple modeling frameworks. The results also show that injury severity in this setting is best understood as a multifactorial outcome shaped by the interaction of crash configuration, impairment, visibility, speed-related behavior, geography, and overall crash intensity. This makes the central contribution of the study clear: the most severe motorcycle injuries in Southern California do not arise from isolated causes, but from recurring patterns of risk that can be identified in routinely collected crash data and used to inform more targeted safety interventions.

5.2 Interpretation in Context of Prior Research

The findings of this study are broadly consistent with the existing motorcycle crash severity literature while also offering context-specific insights for Southern California. Across all three modeling approaches, logistic regression, Random Forest, and XGBoost, a consistent set of behavioral, environmental, and situational predictors emerged as the primary determinants of injury severity. The consistency across modeling frameworks suggests that these relationships reflect underlying crash dynamics rather than model-specific artifacts.

5.2.1 Collision Geometry

Collision geometry, particularly broadside impacts, was one of the most influential factors in predicting injury severity in our study. Broadside collisions accounted for approximately 24.9% of crashes in the dataset and ranked among the strongest predictors in the SHAP analysis (see Table 4.3.1). Prior research has consistently identified crash configuration as an important determinant of injury outcomes, with multi-vehicle interactions producing different severity patterns than single-vehicle events (Wang, 2022).

The present study extends this understanding. In a Southern California context, broadside collisions are both relatively common and consistently associated with more severe outcomes. It is important to distinguish between how often a crash type occurs and how severe it is when it does occur. Descriptive analysis indicated that hit-object and vehicle-pedestrian collisions had higher proportions of fatal outcomes. However, these crash types were less frequent in the dataset. Broadside collisions, by contrast, occurred more often and were consistently associated with elevated severity. As a result, they contributed more strongly to overall model predictions. The SHAP importance rankings reflect this combined effect of frequency and impact rather than fatality rate alone.

5.2.2 Alcohol Involvement

Alcohol involvement showed a clear relationship with injury severity. This is consistent with a well-established body of literature linking impairment to increased crash risk and worse outcomes (Zlatkovic and Zlatkovic, 2022; Burford et al., 2024). In our study, alcohol involvement was recorded in only 7.5% of cases. However, this value must be interpreted cautiously. In SWITRS data, alcohol involvement is only recorded when a test is administered, meaning that many cases are not explicitly coded as either present or absent. To address this limitation, cases with missing values were coded as “unknown,” representing approximately 92.5% of observations. This suggests that the observed proportion of alcohol involvement likely underestimates its true prevalence. Despite this limitation, alcohol involvement consistently appeared as a strong positive contributor to fatal injury predictions across models (see Table 4.3.1), reinforcing its importance as a modifiable behavioral risk factor.

5.2.3 Lighting Conditions

Lighting conditions played a meaningful role as well in injury severity. Crashes occurring in dark environments without street lighting were consistently associated with increased severity in the SHAP analysis (see Table 4.3.1). This pattern has been observed in prior studies as well, with reduced nighttime visibility making it harder to detect hazards in time and increasing the chances of severe or fatal injuries. (Li et al., 2021; Sharafeldin et al., 2022; Wei et al., 2025). The consistency between our study’s data and prior studies suggests that lighting conditions reflect an underlying pattern that influences crash severity rather than a region-specific effect.

5.2.4 Speed-Related Violations

Speed-related violations further contributed to injury severity outcomes. Violations of the Basic Speed Law appeared among the top predictors and were positively associated with fatal injury classification (see Table 4.3.1). This is consistent with prior research demonstrating that higher speeds increase injury severity through greater kinetic energy transfer during collisions (Mamlouk et al., 2024; Rice and Troszak, 2015). In the present dataset, unsafe speed was identified as the primary collision factor in approximately 29.4% of crashes, highlighting both its prevalence and its role in shaping outcomes.

5.2.5 Geographic Variation

Geographic variation also emerged as an important factor. Variables representing crashes occurring within Los Angeles and in unincorporated areas ranked among the top predictors in the SHAP analysis (see Table 4.3.1). Prior research has shown that crash severity relationships can vary across space and time, reflecting differences in roadway design, traffic density, enforcement patterns, and environmental conditions (Li et al., 2021). Although this study does not explicitly model these interactions, the consistent importance of location-based variables suggests that geography plays a meaningful role in shaping injury outcomes in Southern California.

5.2.6 Tow-away as the Strongest Predictor

The tow-away indicator was the strongest predictor in the SHAP analysis and requires careful interpretation. This variable reflects whether a vehicle required towing after a crash. This represents a post-crash outcome rather than a pre-crash condition. Its prominence indicates a strong relationship between vehicle damage and injury severity, but it should not be interpreted as causal. Instead, it serves as a proxy for crash intensity, reinforcing the broader conclusion that higher-energy impacts are more likely to result in severe or fatal injuries (see Table 4.3.1).

5.2.7 Model Performance in Comparison to Prior Studies

Model performance results are consistent with prior findings in the motorcycle safety literature. The macro ROC-AUC values for Random Forest (0.6941) and XGBoost (0.6926) indicate moderate classification performance (see Table 4.2.1). All models struggled to accurately predict the minority fatal injury class (class 1), with recall values ranging from 0.16 to 0.20 (see Table 4.2.2). A consistent misclassification pattern was observed in which fatal and severe injuries were frequently predicted as visible injury (see Table 4.5.2). This suggests the available predictors struggle to clearly separate neighboring severity classes. The consistency of this limitation across modeling approaches implies that the challenge is likely driven by data structure, class imbalance, and the complex nature of injury severity rather than the choice of algorithm.

5.3 Public Safety and Policy Implications

What the models identified points directly toward where policy attention in Southern California is most needed. No single factor drives motorcycle injury severity. The predictors identified here cut across behavior, environment, road conditions, and geography. That spread has real implications for how interventions should be designed. The importance of broadside collision geometry highlights intersections as a critical focus area for intervention. As shown in the model results (see Table 4.3.1), these collisions are among the strongest contributors to severe injury outcomes. Broadside crashes are often associated with right-of-way violations, signal noncompliance, and limited visibility. Targeted intersection safety audits in high-risk areas may help identify opportunities for improvement. Potential countermeasures include enhanced signal timing, improved signage, motorcycle detection systems, and geometric modifications that reduce conflict angles.

Alcohol involvement remains one of the most significant behavioral risk factors identified in this study. Even with extensive underreporting, it consistently appeared as a strong predictor of fatal injury (see Table 4.3.1). Sobriety checkpoints concentrated in areas with heavy motorcycle traffic, especially after dark, remain a practical response. Public education focused on impaired riding could help fill a gap, since most current messaging is aimed at drivers rather than motorcyclists.

Environmental conditions, lighting in particular, represent another important area for intervention. The association between dark, unlit roadways and increased injury severity (see

Table 4.3.1) suggests that improving lighting infrastructure could reduce the severity of nighttime crashes. Incorporating motorcycle crash data into roadway lighting prioritization strategies may help identify high-risk corridors, particularly in unincorporated areas where lighting infrastructure is often limited.

Speed management is another key policy consideration. With unsafe speed identified as the primary collision factor in nearly one-third of crashes, and as a meaningful predictor in the model (see Table 4.3.1), interventions aimed at reducing speed may have substantial safety benefits. Options worth considering include stricter enforcement on known high-speed corridors and design changes that make lower speeds the path of least resistance.

The spatial patterns observed in our study suggest that safety interventions should be geographically targeted rather than uniformly applied. The importance of location-based predictors (see Table 4.3.1) indicates that risk is not evenly distributed across the region. Unincorporated areas face a unique combination of challenges. Challenges like faster roads, fewer infrastructure protections, and less consistent enforcement. This makes them strong candidates for broad, coordinated safety efforts that span road design, policing, and emergency preparedness. In contrast, areas like Los Angeles may benefit from interventions that improve rider visibility, raise driver awareness, and better address how motorcycles and other vehicles interact in dense traffic.

This study highlights the value of combining traditional crash data with machine learning and interpretability techniques. Using SHAP analysis made it possible to look beyond overall model accuracy and examine which conditions were driving predictions toward more severe outcomes (see Table 4.3.1). This approach builds on traditional crash frequency analysis and can help transportation agencies prioritize interventions, allocate resources more effectively, and assess safety strategies over time.

5.4 Study Limitations

The study relies on police-reported crash data from California's Statewide Integrated Traffic Records System (SWITRS) and DMV registration data. A major limitation is the risk of reporting bias, misclassification, and missing information. For instance, injury severity is assessed using the KABCO scale. This scale depends on evaluations by law enforcement at the scene, not clinical medical records, which may distort long-term outcomes.

Unobserved rider-related risk factors, such as experience and helmet quality, are absent from police data. These factors must be inferred, potentially introducing bias. The dataset focuses solely on acute injury severity at the crash moment. This method neglects long-term health effects or delayed fatalities.

Another challenge is missing data, especially regarding behavioral indicators. Alcohol involvement is marked as "Unknown" in 92.5% of cases, a result of "Missing Not At Random" (MNAR) practices. Additionally, variables like ramp intersections and violation subsections are often omitted. This complicates standard imputation.

5.5 Methodological Constraints

This study employs retrospective observational design. This inherently limits causal claims. Although methods such as Multinomial Logistic Regression and XGBoost identify strong associations, they cannot confirm causal links. Thus, all results should be viewed as associative. A significant challenge lies in the extreme class imbalance within the KABCO distribution. Fatal outcomes make up only 4.9% of the dataset, while "Visible Injuries" account for 44.9%. Even with adjustments for class weight and the use of macro-averaged F1 scores, modeling rare fatal outcomes remains difficult.

The modeling approach also faces issues with dimensionality. One-hot encoding for categorical variables expands the feature space significantly, up to 472 variables. This can heighten sensitivity to noise and obscure the interpretability of importance rankings in tree-based models. While techniques like XGBoost have strong predictive power, they may lack analytical traceability compared to traditional regression.

5.6 Generalizability Considerations

The generalizability of these findings is constrained by their specific geographic and cultural focus. The study examines five Southern California counties: Los Angeles, San Diego, Riverside, Orange, and San Bernardino. This region features dense urban stop-and-go traffic, a mellow riding climate, rural and temperate mountainous roadway, and somewhat uncommon legal practices like lane-splitting. Such elements may not reflect conditions in other areas. Therefore, the findings may not translate well to regions with different traffic patterns, enforcement practices, or riding cultures. Factors affecting severity in an urban area like Los Angeles could differ widely in inclement weather settings or areas with limited thoroughfare infrastructure. Moreover, predictor effects such as nighttime travel or helmet efficacy might change over time due to sensors, autonomy, and material sciences, suggesting that relationships identified in this dataset could change given time.

5.7 Recommendations for Future Research

Future research should build on the present study by addressing the limitations of police-reported crash data and expanding the range of variables used to model motorcycle injury severity. Although this study identified meaningful relationships across behavioral, environmental, roadway, spatial, and temporal domains, the available data still lacked direct measures of rider experience, helmet quality, risk perception, aggressiveness, and situational decision-making. These omissions limit the ability to fully explain why similar crash contexts sometimes lead to different injury outcomes. Future studies should therefore incorporate richer rider-level and crash-context variables, whether through linked medical data, licensing histories, trauma outcomes, helmet use detail, emergency response measures, or survey-based rider behavior information. Expanding the data structure in this way would improve both explanatory depth and predictive precision.

A second recommendation is to extend the geographic and temporal scope of analysis. This study focused on Southern California because the region presents a distinct traffic and exposure environment shaped by dense urban networks, high motorcycle volume, and varied roadway contexts. That regional focus strengthened the study’s practical relevance, but it also limits broader generalizability. Future research should test whether the same predictors remain important across other California regions or in multistate comparisons with different riding cultures, roadway systems, and enforcement conditions. Longitudinal designs would be especially valuable for examining whether the effects of lighting, speed-related behavior, alcohol involvement, and collision geometry remain stable over time or shift across changing roadway and policy conditions.

Future work should also place greater emphasis on spatial and spatiotemporal modeling. The present study found that geography mattered, with location-based variables contributing meaningfully to model output, yet the analysis did not fully model spatial interaction effects or temporal instability. Prior literature reviewed in this thesis suggests that injury severity relationships are not constant across time or place, and the present findings support the value of pursuing that direction more directly. Geographically weighted models, hierarchical spatial models, and spatiotemporal machine learning approaches may help identify localized clusters of risk and determine whether certain predictors are stronger in specific counties, roadway environments, or time periods.

Another important direction is the refinement of exposure measurement. This study improved on raw crash-count approaches by incorporating county-level motorcycle registration data as an exposure variable, but registration counts remain a broad proxy rather than a direct measure of actual riding exposure. Future studies should incorporate more precise measures such as annual motorcycle miles traveled, corridor-level traffic counts, roadway-specific motorcycle volume, commute versus recreational riding patterns, or seasonal exposure variation. Better exposure measurement would help distinguish true risk from simple differences in rider concentration and would improve interpretation of geographic disparities in crash severity.

Methodologically, future research should continue exploring approaches that improve performance on rare but high-consequence outcomes. In this study, all three models struggled with the minority fatal injury class despite stable overall generalization, highlighting the difficulty of predicting the most severe cases from imbalanced observational data. Future work should examine alternative strategies for rare-event classification, including more advanced resampling procedures, cost-sensitive learning, calibrated ensemble methods, and hybrid modeling frameworks that combine interpretable statistical models with higher-capacity machine learning systems. At the same time, interpretability should remain a priority so that improved prediction does not come at the expense of practical usability for transportation safety decision-making.

Finally, future research should more directly investigate interaction effects that are strongly suggested by both the literature and the present results. The findings of this study indicate that motorcycle injury severity is multifactorial and context dependent, meaning that individual predictors may operate differently when combined with other conditions such as nighttime travel, roadway type, unincorporated geography, or alcohol involvement. Future studies should therefore move beyond isolated variable importance and test how combinations of behavioral, environmental, and roadway factors jointly shape severe and fatal injury outcomes. This would provide a more realistic understanding of crash severity risk and could help support more targeted and context-specific intervention strategies.

References

Adeyemi, O., Paul, R., Delmelle, E., DiMaggio, C., & Arif, A. (2023). Road environment characteristics and fatal crash injury during the rush and non-rush hour periods in the U.S: Model testing and cluster analysis. *Spatial and Spatio-temporal Epidemiology, 44*, 100562.

Agyemang, W., Adanu, E. K., & Jones, S. (2021). Understanding the factors that are associated with motorcycle crash severity in rural and urban areas of Ghana. *Journal of Advanced Transportation, 2021*, Article 6336517.

Beeharry, R., Ratanavaraha, V., & Goodary, R. (2021). Road engineering determinants of moped and motorcycle crashes at non-intersection road segments in a developing country - Mauritius. *International Journal of Civil Infrastructure, 4*, 85-98.

Burford, K. G., Rundle, A. G., Frangos, S., Pfaff, A., Wall, S., Adeyemi, O., & DiMaggio, C. (2024). Comparing alcohol involvement among injured pedalcycle and motorcycle riders across three national public-use datasets. *Traffic Injury Prevention, 25*(8), 1023-1030.

Champahom, T., Se, C., Aryuyo, F., Banyong, C., Jomnonkwao, S., & Ratanavaraha, V. (2023). Crash severity analysis of young adult motorcyclists: A comparison of urban and rural local roadways. *Applied Sciences, 13*, 11723.

Chang, H.-L., & Yeh, T.-H. (2007). Motorcyclist accident involvement by age, gender, and risky behaviors in Taipei, Taiwan. *Transportation Research Part F: Traffic Psychology and Behaviour, 10*(2), 109-122.

Cheng, W., Gill, G. S., Sakrani, T., Dasu, M., & Zhou, J. (2017). Predicting motorcycle crash injury severity using weather data and alternative Bayesian multivariate crash frequency models. *Accident Analysis & Prevention, 108*, 172-180.

Farid, A., & Ksaibati, K. (2021). Modeling severities of motorcycle crashes using random parameters. *Journal of Traffic and Transportation Engineering (English Edition), 8*(2), 225-236.

Haque, M. M., Chin, H. C., & Lim, B. C. (2008). The influence of motorcyclist behavior in crash involvement. In *Proceedings of the 4th International Conference on Traffic and Transport Psychology*.

Hsu, T.-P., & Wen, K.-L. (2022). Using multinomial regression to explore the spatial factors affecting left-turn oncoming accidents involving motorcycles. *Traffic Injury Prevention, 23*(1), 46-50.

Ijaz, M., Liu, L., Almarhabi, Y., Jamal, A., Usman, S. M., & Zahid, M. (2022). Temporal instability of factors affecting injury severity in helmet-wearing and non-helmet-wearing motorcycle crashes: A random parameter approach with heterogeneity in means and variances. *International Journal of Environmental Research and Public Health, 19*, 10526.

Islam, M., & Li, X. (2026). Exploring the motorcycle crash risks and riders' risk profiles: Evidence from the motorcycle crash causation study. *IATSS Research, 50*, 691-700.

Jafari, M., Starewich, M., Hossain, A., Barua, S., Alnawmasi, N., Ye, X., & Das, S. (2025). Assessing motorcyclist injury severity on curved road segments with temporal dynamics and unobserved heterogeneity. *Scientific Reports, 15*, 97972.

Li, X., Liu, J., Zhang, Z., Parrish, A., & Jones, S. (2021). A spatiotemporal analysis of motorcyclist injury severity: Findings from 20 years of crash data from Pennsylvania.

Accident Analysis & Prevention, 151, 105952.

Lin, M.-R., Chang, S.-H., Huang, W., Hwang, H.-F., & Pai, L. (2003). Factors associated with severity of motorcycle injuries among young adult riders. *Annals of Emergency Medicine, 41*(6), 783-791.

Mamlouk, A., Teymouri, A., & Ghodrat Abadi, M. (2024). Motorcycle crashes in California: Analysis of crash severity and contributing factors. *American Society of Civil Engineers (ASCE).*

Mustakim, F., Ahmad, M. N., Kadir, R. A., Charlton, S. G., Aziz, A. A., et al. (2023). Sustainable hydrodynamic of artificial neural networks and logistic regression model to lane change serious conflict at unsignalized intersection on Malaysia's Federal Route. *Journal of BioMed Research and Reports, 2*(5), 1-16.

Naseralavi, S., Soltanirad, M., Igene, M., Jimee, K., & Ranjbar, E. (2025). Determining factors affecting motorcycle crash severity based on time of day and season combinations. *ENG Transactions, 6*, Article 2941.

National Highway Traffic Safety Administration. (2024). *Motorcycles: 2022 data (Traffic Safety Facts. Report No. DOT HS 813 589).* National Highway Traffic Safety Administration.

National Highway Traffic Safety Administration. (2025). *Motorcycles: 2023 data (Traffic Safety Facts. Report No. DOT HS 813 732).* National Highway Traffic Safety Administration.

Ospina-Mateus, H., Quintana Jiménez, L. A., Lopez-Valdes, F. J., Garcia, S. B., Barrero, L. H., & Sana, S. S. (2021). Prediction of motorcycle traffic crashes in Cartegena: Development of a safety performance function. *Journal of Ambient Intelligence and Humanized Computing*, 1264-1278.

Rezapour, M., Farid, A., Nazneen, S., & Ksaibati, K. (2021). Using machine learning techniques for evaluation of motorcycle injury severity. *IATSS Research, 45*(2), 277-285.

Rezapour, M., Nazneen, S., & Ksaibati, K. (2020). Application of deep learning techniques in predicting motorcycle crash severity. *Engineering Reports, 2*, e12175.

Safe Transportation Research and Education Center (SafeTREC). (2025). 2025 SafeTREC Traffic Safety Facts: Motorcycle Safety. University of California, Berkeley.

Santos, K., Firme, B., Dias, J. P., & Amado, C. (2021). Analysis of motorcycle accident injury severity and performance comparison of machine learning algorithms. *Transportation Research Record, 2678*(1), 737-748

Sharafeldin, M., Farid, A., & Ksaibati, K. (2022). Investigating the impact of roadway characteristics on intersection crash severity. *Eng, 3*, 418-422.

Smith, M. C., Stonko, D. P., Guillamondegui, O. D., & Dennis, B. M. (2018). Temporal factors drive motorcycle collision-related trauma. *The American Surgeon, 84*(9), e363-e365.

Tian, L., & Zhang, C. (2024). The effect of age on motorcyclist injury severities in multi-vehicle motorcycle crashes: Accounting for unobserved heterogeneity and insights from out-ofsample prediction. *IEEE Access.*

Wahab, L., & Jiang, H. (2019). A comparative study on machine learning based algorithms for prediction of motorcycle crash severity. *PLoS ONE, 14*(4), e0214966.

Wang, M.-H. (2022). Investigating the difference in factors contributing to the likelihood of motorcyclist fatalities in single motorcycle and multiple vehicle crashes. *International Journal of Environmental Research and Public Health, 19*, 8411.

Wei, F., Zhou, Y., Guo, Y., Wang, D., Han, H., Guo, D., & Pirov, J. (2025). Exploring the impact of curve and slope on the motorcycle crash severity on rural roads. *Journal of Transportation Safety & Security, 17*(12), 1480-1502.

Wei, T., Zhu, T., Lin, M., & Liu, H. (2024). Predicting and factor analysis of rider injury severity in two-wheeled motorcycle and vehicle crash accidents based on an interpretable machine learning framework. *Traffic Injury Prevention, 25*(2), 194-201.

Zlatkovic, M., & Zlatkovic, S. (2022). Multinomial logistic regression modeling of motorcycle crash severities - Wyoming case study. *Journal of Road and Traffic Engineering, 68*(2).

Appendix A: Analysis of Predictors Across Several Key Studies

Statistically Significant Predictors	Strongest Predictors (High Effect Size/Importance)	Conflicting Findings across Studies	In-Text Citations & References (APA 7)
Alcohol Involvement: BAC levels consistently correlate with increased fatality and hospitalization	Rural Alcohol Use: In rural settings, alcohol involvement showed an extreme Odds Ratio (OR) of 44.68 for fatal outcomes	Gender: While most studies identify male riders as higher risk, one model found a negative correlation (-2.875) between male gender and severity	Naseralavi, S., et al. (2025). Determining factors affecting motorcycle crash severity based on time of day and season combinations. ENG Transactions.
Speed Violations: Exceeding speed limits or safe speeds for conditions	Unlicensed Riding: Being unlicensed increased the marginal probability of a fatal outcome by 14.75% on urban roadways.	Helmet Efficacy: Overwhelmingly protective across studies (OR 5.87 for fatality), yet one study on young riders found helmet use statistically insignificant in their specific local roadway model.	Champahom, T., et al. (2023). Crash severity analysis of young adult motorcyclists: A comparison of urban and rural local roadways. Applied Sciences
Rider Age: Risk profiles vary significantly by age, with younger and older riders being most vulnerable.	Younger Age: Motorcyclists aged 16–25 were found to have an OR of 11.84 for crash involvement compared to other groups.	Weather Conditions: Clear weather is generally safer for visibility, but some data suggest it correlates with increased crash frequency due to higher travel speeds and rider complacency	Islam, M., & Li, X. (2026). Exploring the motorcycle crash risks and riders' risk profiles: Evidence from the motorcycle crash causation study. IATSS Research
Road Alignment: Horizontal curves and vertical slopes significantly impact control	Barrier Collisions: Impacting a fixed barrier in a single-vehicle crash resulted in an average OR of 6.79 for severe injury	Time of Day: Morning peak hours generally have lower crash counts but often exhibit a higher case fatality rate (CFR) than afternoon peaks	Farid, A., & Ksaibati, K. (2021). Modeling severities of motorcycle crashes using random parameters. Journal of Traffic and Transportation Engineering,

Collision Type: Head-on, broadside, and fixed-object collisions are frequent predictors of high severity.	Speed Limits: Posted speed limits ≥ 70 mph were among the strongest predictors of fatality, with an OR of 3.92.	Traffic Volume (AADT): High traffic is often associated with more crashes, but some models show it can reduce severity due to naturally lower speeds in congestion.	Wang, M.-H. (2022). Investigating the difference in factors contributing to the likelihood of motorcyclist fatalities in single motorcycle and multiple vehicle crashes. International Journal of Environmental Research and Public Health

Predictors to Effect Size, Odds Ratios, and Importance Scores Across Known Studies

Predictor	Effect Size / Odds Ratio (OR) / Importance Score	Dependent Variable	Source (APA 7)
Alcohol Involvement (BAC .08+)	OR: 4.527 1	Motorcyclist Fatality	Wang (2022)
Speed Limit (PSL ≥ 70 mph)	OR: 3.920 1	Motorcyclist Fatality	Wang (2022)
Younger Rider (< 25 years)	OR: 0.604 1	Motorcyclist Fatality	Wang (2022)
Location Type	Importance: 0.12236 (Gain Ratio)	Motorcycle Crash Severity	Wahab & Jiang (2019)
Settlement Type	Importance: 0.05267 (Gain Ratio)	Motorcycle Crash Severity	Wahab & Jiang (2019)
Helmet Usage	β: -0.661 (Marginal Effect: -1.82% for Fatal)	Injury Severity (KABCO)	Li et al. (2021)
Truck Involvement	β: 0.287 (Marginal Effect: +0.93% for Fatal)	Injury Severity (KABCO)	Li et al. (2021)
Motorcyclist Alcohol (BAC)	SHAP Importance: ~0.34	Injury Severity	Santos et al. (2024)
Road Type	SHAP Importance: ~0.12	Injury Severity	Santos et al. (2024)
Collision with Fixed Object	Marginal Effect: +2.60% (for Fatal)	Injury Severity	Jafari et al. (2025)
Daylight	Marginal Effect: -2.52% (for Fatal)	Injury Severity	Jafari et al. (2025)

Unlicensed Rider	Marginal Effect: +14.75% (for Fatal)	Fatality (Urban Roadways)	Champahom et al. (2023)
Exceeding Speed Limit	Marginal Effect: +11.00% (for Fatal)	Fatality (Urban Roadways)	Champahom et al. (2023)
No. of Vehicles (3 Vehicles)	β: 1.072	Crash Severity (Injury/Fatality)	Naseralavi et al. (2025)
Road Surface (Snowy, Icy, Wet)	β: 0.932	Crash Severity (Injury/Fatality)	Naseralavi et al. (2025)
Old Driver (Age > 51)	Marginal Effect: +2.4% (for SFI)	Injury Severity (Rural Roads)	Wei et al. (2025)
Head-on Collision	Marginal Effect: +1.2% (for SFI)	Injury Severity (Rural Roads)	Wei et al. (2025)
Younger Rider (16-25)	OR: 11.846	Crash Involvement	Islam & Li (2026)
Licensed Rider	OR: 0.263	Crash Involvement	Islam & Li (2026)
Pavement Friction (FN40R)	Marginal Effect: -0.36% (for KA)	Intersection Crash Severity	Sharafeldin et al. (2022)
Urban Location	Marginal Effect: -1.14% (for KA)	Intersection Crash Severity	Sharafeldin et al. (2022)
Red Light Running (Afternoon)	Marginal Effect: +6.5%	Traffic Violations	Fu & Liu (2020)
Roll Acc. peak-to-RMS	AUC: 0.996	Loss of Control Prediction	Huertas-Leyva et al. (2020)
Steering Rate Max	AUC: 0.980	Loss of Control Prediction	Huertas-Leyva et al. (2020)
Barrier Crash (Single Vehicle)	Average OR: 6.794	Severe Crash	Farid & Ksaibati (2021)
RUI (Single Vehicle)	Average OR: 4.527	Severe Crash	Farid & Ksaibati (2021)
Wet Road Surface	Mean β: 0.341	Crash Severity	Mamlouk et al. (2024)
Failure to Yield	β: 0.394	Crash Severity	Mamlouk et al. (2024)
Traffic Speed (60+ MPH)	Risk Ratio: 1.25	Torso Injury (Lane Splitting)	Rice & Troszak (2015)
Speed Differential (15+ MPH)	Risk Ratio: 1.58	Torso Injury (Lane Splitting)	Rice & Troszak (2015)
Age < 20	OR: 4.484	Accident Occurrence	Chang & Yeh (2007)

Violation Facet (High Risk)	OR: 1.702	Accident Occurrence	Chang & Yeh (2007)
Alcohol Involvement (Rural)	OR: 44.6847 (for Fatal)	Crash Severity	Zlatkovic & Zlatkovic (2022)
Helmet (None Used)	OR: 5.8717 (for Fatal)	Crash Severity	Zlatkovic & Zlatkovic (2022)
Kilometers Traveled/Year	SHAP Importance: ~1.2	Rider Injury Severity	Wei et al. (2024)
Throwing Distance (> 1000 cm)	SHAP Importance: ~0.8	Fatal Injury	Wei et al. (2024)
Precipitation (Compl. of Pain)	β: -0.389	Crash Frequency	Cheng et al. (2017)
No. of Accesses (Intersections)	β: 0.30	Motorcycle Crashes / km	Ospina-Mateus et al. (2021)
Crash Time (00:00-04:59)	OR: 4.787	Severe Injury	Beeharry et al. (2021)
Interstate Road	IRR: 1.82	Fatal Crash Injury Counts	Adeyemi et al. (2023)
Rainy Weather	IRR: 1.10	Fatal Crash Injury Counts	Adeyemi et al. (2023)
Lane Change Angular Conflict	Importance: 0.332 (ANN)	Serious Conflict	Mustakim et al. (2023)
Distraction (Helmet Wearing)	Marginal Effect: +3.82% (for FI)	Injury Severity (2017)	Ijaz et al. (2022)
Injury-to-Admission Time	OR: 5.63	Mortality (TBI)	Nugroho & Masrika (2025)
Risk-Taking Behavior (High)	OR: 2.21	Crash Vulnerability	Haque et al. (2008)

Appendix B: Model Variable Dictionary for Retained Predictor Variables

Temporal Variables

Definition: These describe *when* the crash occurred.

TIME_OF_DAY_Night

Definition: Crash occurred during nighttime hours (TIMS uses 9 PM-5 AM).

Full Time-of-Day Classes:

- Morning
- Afternoon
- Evening
- Night

Environmental Conditions

Definition: These describe lighting and visibility conditions.

LIGHTING_A - Daylight

Definition: Crash occurred during daylight.

LIGHTING_D - Dark, Street Lights On

Definition: Crash occurred at night with functioning street lighting.

Full Lighting Classes:

- A - Daylight
- B - Dusk/Dawn
- C - Dark, Street Lights Off
- D - Dark, Street Lights On
- E - Dark, No Street Lights
- F - Other
- G - Not Stated

Roadway Characteristics

Definition: These describe collision configuration, roadway context, and vehicle types.

TYPE_OF_COLLISION_B - Broadside

Definition: Front of one vehicle strikes the side of another ("T-bone").

TYPE_OF_COLLISION_D - Sideswipe, Same Direction

Definition: Vehicles travelling in the same direction scrape or collide along their sides.

TYPE_OF_COLLISION_E - Sideswipe, Opposite Direction

Definition: Vehicles travelling in opposite directions scrape or collide along their sides.

Full Collision Type Classes:

A Head-On; **B** Broadside; **C** Rear-End; **D** Sideswipe (Same Direction); **E** Sideswipe (Opposite Direction); **F** Hit Object; **G** Overturned; **H** Vehicle/Pedestrian; **I** Other; **J** Not Stated.

CITY_UNINCORPORATED - Unincorporated Area

Definition: Crash occurred outside incorporated city limits.

Full City Classes:

- Named incorporated cities
- Unincorporated

STWD_VEHTYPE_AT_FAULT_C - Motorcycle at Fault

Definition: Standardized TIMS vehicle type indicates the at-fault party was riding a motorcycle.

Full Standardized Vehicle Type Classes:

A Passenger Car; **B** Pickup/Panel Truck; **C** Motorcycle; **D** Truck/Tractor; **E** Bus; **F** Emergency Vehicle; **G** Bicycle; **H** Pedestrian; **I** Other; **J** Unknown.

CHP_VEHTYPE_AT_FAULT_02 - Passenger Car at Fault

Definition: CHP vehicle type code 02 = passenger car.

Common CHP Vehicle Type Codes:

01 Motorcycle; **02** Passenger Car; **03** Pickup; **04** Van; **05** Truck; **06** Tractor/Trailer; **07** Bus; **08** Emergency Vehicle; **09** Bicycle; **10** Pedestrian; 11+ Other/Unknown.

TOW_AWAY_Y - Vehicle Towed

Definition: At least one involved vehicle required towing due to damage.

Full Tow-Away Classes:

- Y - Yes
- blank - No

Behavioral Indicators

Definition: These describe human behavior, violations, impairment, and unsafe actions.

PCF_VIOL_CATEGORY_01 - DUI

Definition: Primary collision factor was alcohol or drug impairment.

PCF_VIOL_CATEGORY_07 - Unsafe Speed

Definition: Primary collision factor was traveling too fast for conditions or exceeding the posted limit.

Full PCF Violation Categories:
01 DUI; **02** Impeding Traffic; **03** Unsafe Speed; **04** Following Too Closely; **05** Wrong Side of Road; **06** Improper Passing; **07** Unsafe Lane Change; **08** Improper Turning; **09** Auto Right-of-Way; **10** Pedestrian Right-of-Way; **11** Pedestrian Violation; **12** Signals/Signs; **13** Hazardous Parking; **14** Lights; **15** Brakes; **16** Other Equipment; **17** Other Hazardous Violation; **18** Other Than Driver; **19** Unknown; **20** Fell Asleep.

PCF_VIOLATION_21658.0 - Unsafe Lane Change (CVC 21658)

Definition: Failure to maintain a single lane or unsafe lateral movement.

PCF_VIOLATION_23152.0 , DUI Alcohol (CVC 23152)

Definition: Driving under the influence of alcohol.

Full CVC Violation Classes:
Hundreds of California Vehicle Code sections; dataset includes only those present in the crashes.

ALCOHOL_INVOLVED_Y - Alcohol Involved

Definition: At least one party had been drinking.

Full Alcohol Classes:

- Y - Yes
- blank - No

HIT_AND_RUN_M - Misdemeanor Hit and Run

Definition: Driver fled the scene; offense classified as misdemeanor.

Full Hit-and-Run Classes:

- F - Felony
- M - Misdemeanor
- blank - Not Hit-and-Run

Appendix C: Analytical Code and Modeling Workflow